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Individual Report - “Physio-Connect”

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Declaration Page

I hereby declare that this project report and all the artifacts associated is my original work, completed as part of my academic requirements,and has not been previously submitted for any other academic program.

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Abstract + Keywords

This report presents a comprehensive three-chapter framework for system development. Chapter 4, the System Requirements Specification (SRS), delves into a Stakeholder Analysis, Requirement Elicitation Techniques, and Sequence Diagrams, along with a detailed specification of functional and non-functional requirements. Chapter 5 explores the Social, Legal, Ethical, and Professional (SLEP) issues, addressing potential concerns and proposing effective mitigation strategies to ensure compliance and ethical integrity. Finally, Chapter 6 focuses on System Architecture and Design, incorporating detailed design methodologies, technology reviews, and visual representations of the system architecture. This structured approach ensures a holistic understanding of user needs, regulatory considerations, and technical design, laying the foundation for an efficient and user-centered system

Acknowledgement

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Chapter 4: System Requirements Specification (SRS)

4.1 Chapter Overview

The System Requirements specification (SRS) for the proposed physiotherapy (Physio-connect) platform. It begins with a comprehensive stakeholder analysis to identify the individuals and groups who will interact with and influence the development and deployment of the app. Stakeholder perspectives are gathered through interviews and surveys, which inform the requirements elicitation process. Following this, the chapter will outline the system's use cases and provide a thorough breakdown of both functional and nonfunctional requirements essential for delivering a seamless user experience.

4.2. Stakeholder Analysis

To ensure that physio-connect aligns with the needs of all relevant parties, a stakeholder analysis is conducted. This analysis identifies key stakeholders, highlights their roles, and assesses their impact on the development and use of the app. The Onion model diagram (below) visually represents the primary, secondary, and tertiary stakeholders in relation to the core system.

4.2.1.Onion Model

The Onion Model illustrates the hierarchical structure of stakeholders, categorizing them into three layers:the wider environment, the containing system, and the system itself, based on their level of interaction and influence. A detailed diagram representing this model is provided on the following page.

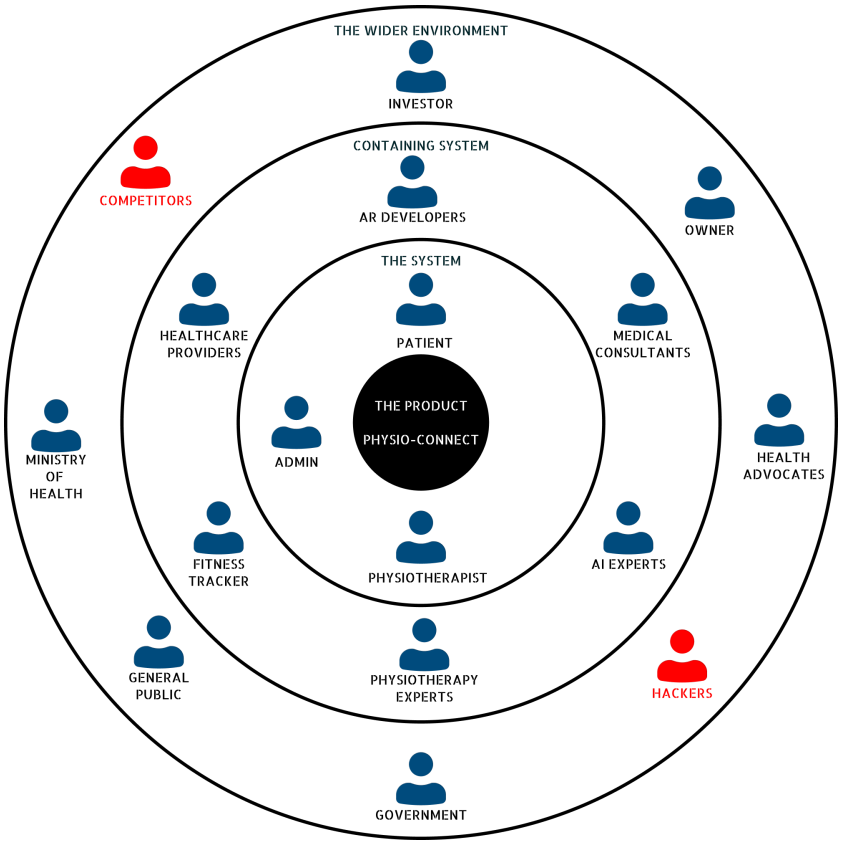


Figure 01 : The Onion Module (Self Composed)

4.2.2.Stakeholder Descriptions

Stakeholder	Viewpoint
Functional beneficiary	
Users (Patients, Athletes)	Will use app's features, such as guided exercises, tracking, and recovery insights, to manage and accelerate their recovery process.
Physiotherapist	Will offer remote assistance, monitor patients progress, and broaden their reach.

Financial beneficiary	
Platform Owner	Will make money by getting more patients / Clients.
Investors	Will make money through profits, when the system is being used by more people.
Social beneficiary	
General Public	Will benefit from a healthier society, as the app reduces barriers to accessing physiotherapy and promotes overall physical well-being.
Health Advocates	Can leverage the app to promote physical health awareness and encourage injury prevention and rehabilitation.
Families of the users	Will gain peace of mind knowing their loved ones have access to convenient and reliable physiotherapy resources, reducing recovery stress.
Operational beneficiary	
Physiotherapists	Will benefit operationally by accessing a user-friendly system to manage multiple patients, track their progress, and optimize treatment plans remotely.
Medical Professionals	Will have a platform to share insights, collaborate on complex cases, and ensure patients receive effective care.
Health Care Providers	Will benefit from partnerships with the app, using it as a resource for patient rehabilitation and wellness programs.
Negative Stakeholders	
Competitors	May lose market share or clientele to the app due to its innovative features and ease of access.
Hackers	Could exploit vulnerabilities to access sensitive health data, posing a security threat to the platform.
Regulatory	
Ministry of health	Will ensure the app complies with healthcare standards, regulations, and ethical practices.
Government	Will oversee the platform to ensure it aligns with public health

	objectives and data privacy laws.
Experts	
AR developers	Will enhance the platform's Augmented Reality features, ensuring seamless and accurate delivery of visual guides for exercises and posture correction.
Physiotherapy experts	Will contribute to the app's content, ensuring medical accuracy in recommendations, exercises, and recovery strategies.
AI specialists	Will optimize algorithms for personalized recommendations, chatbots, and progress tracking, ensuring an intuitive and adaptive user experience.
Neighboring systems	
Fitness tracker	Will sync with the app to provide users with real-time physical activity data, enhancing insights into recovery progress and daily activity levels.

4.3. Selection of Requirement Elicitation Techniques/Methods

4.3.1. Interviews

Interviews are a direct and effective technique for gathering requirements by engaging stakeholders in meaningful dialogue. This method was implemented to ensure that the Physio Connect app meets its objectives by addressing the needs, expectations, and concerns of all relevant stakeholders. (Software, 2021)

The steps undertaken are outlined below:

Stakeholder Identification

Identifying all critical stakeholders, including patients, physiotherapists, and healthcare administrators, was prioritized. Engaging these groups early allowed the team to gain their insights, address their needs, and resolve potential concerns. This collaborative

approach ensures the app's functionality aligns with user expectations and builds trust, fostering long-term adoption.

Interview Preparation

Careful planning was key to conducting successful interviews. The team defined clear objectives, created structured question frameworks, and set up schedules to ensure interviews were focused, efficient, and productive. Setting expectations and selecting the right participants further contributed to effective data collection.

One-on-One Discussions

Personal interviews were conducted to gain in-depth insights into individual stakeholder requirements:

- Patients: Gathered feedback on app usability, virtual consultation preferences, and personalized features.
- Physiotherapists: Identified workflow needs, such as managing client sessions, tracking progress, and accessing medical histories.

Group Interviews

Group discussions encouraged a collaborative environment for stakeholders to share diverse perspectives and generate innovative ideas.

- Patients: Shared common challenges and suggested desired features, fostering shared solutions.
- Physiotherapists and Healthcare Teams: Brainstormed on features like progress tracking and integration with existing healthcare systems to enhance efficiency.

Post-Interview Analysis

The responses were analyzed to identify trends, recurring challenges, and user priorities, helping the development team refine app features and functionalities.

4.3.2 Surveys and Questionnaires

Surveys and questionnaires were essential tools for collecting quantitative and qualitative data from a broader audience. They provided insights into user preferences, behaviors, and requirements for Physio Connect's development. (R. Feaster, n.d.)

Objective Definition

Establishing specific goals ensured the surveys collected actionable and relevant data. These objectives focused on uncovering user challenges, preferred features, and potential areas for improvement, enabling the team to align the app's features with real-world needs.

Selecting Participants

A diverse respondent pool was selected to capture comprehensive input. Patients, physiotherapists, healthcare administrators, and technology enthusiasts were included to ensure a range of perspectives, addressing both functional and usability needs.

Questionnaire Design

Questions were crafted to be clear, concise, and purposeful, capturing insights into app usability, user satisfaction, and desired innovations. Questions ranged from multiple-choice to open-ended formats, ensuring both quantitative metrics and qualitative insights were obtained.

Survey Distribution

Surveys were distributed via various channels, including:

- Social media platforms to maximize reach.
- Email campaigns targeting healthcare professionals and patients.
- Data Collection and Analysis

The feedback was compiled and analyzed to identify trends, common pain points, and feature requests. Quantitative results were used to prioritize functionalities, while qualitative responses informed design refinements and user experience enhancements.

4.3.3. Prototyping

Prototyping involves creating an initial version of the system to evaluate its design and functionality before full-scale development. It enables stakeholders to visualize the app's features, identify usability issues, and provide constructive feedback.

Implementation of Prototyping:

- **UI/UX Design**: A prototype for **Physio Connect** was created using Figma, showcasing the app's layout, features, and user flow.
- **Iterative Feedback Cycles**: Stakeholders reviewed the prototype to ensure it met their needs. For example, physiotherapists requested modifications to the session-tracking interface to streamline workflow, which were incorporated into subsequent iterations.
- **Low and High-Fidelity Prototypes**: Early low-fidelity wireframes focused on basic layout and user flow, while high-fidelity versions added detailed visuals and interactive elements, simulating the final product's functionality.

Prototyping not only accelerated the development process but also ensured that the app design aligns with user expectations.

4.3.4. Use Case Scenarios

Use case scenarios outline how users interact with a system to achieve specific goals, providing a structured understanding of user workflows. These scenarios detail the steps taken by different users, such as patients, physiotherapists, and administrators, highlighting the system's functionalities and identifying areas for improvement. For example, a use case might describe

how a patient logs in, accesses guided exercises, and tracks recovery progress, while a physiotherapist uses the system to monitor patient progress and update treatment plans. By addressing user behavior and potential challenges, these scenarios help align system design with user needs and improve overall usability.

4.3.5 Focus Groups

Focus groups involve moderated discussions with a select group of participants to gain insights into their preferences, experiences, and expectations. This method provided qualitative data to inform the design and development of Physio Connect.

Implementation of Focus Groups:

- **Target Groups:** Separate sessions were conducted for patients and physiotherapists to understand their distinct needs. For example, patients highlighted the importance of easy navigation and visual feedback during exercises, while physiotherapists emphasized the need for robust reporting tools.
- **Scenarios and Mockups:** Participants were shown prototype mockups and asked to evaluate potential features such as progress tracking, alerts, and in-app communication. This feedback directly influenced feature prioritization.
- **Collaborative Problem Solving:** The focus groups facilitated brainstorming, with participants suggesting solutions to common challenges like integrating wearable device data for better recovery tracking.

By leveraging the diverse perspectives from these sessions, the development team ensured the app caters to real-world requirements.

4.3.6 Document Analysis

Document analysis involves reviewing existing resources, such as technical manuals, policies, and workflows, to extract valuable insights and ensure compliance. This technique was particularly useful in identifying gaps and aligning the app with industry standards.

Application of Document Analysis:

- **Existing Physiotherapy Practices:** Manuals and protocols from physiotherapy clinics were analyzed to understand current practices. For example, the team examined session management techniques and rehabilitation progress tracking to design app features that replicate or enhance these workflows.
- **Regulatory Compliance:** Guidelines from health authorities were reviewed to ensure the app adheres to legal and ethical standards, particularly concerning patient data privacy and security.
- **Competitive Analysis:** Documentation from existing fitness and health apps was studied to identify strengths and weaknesses, enabling Physio Connect to address market gaps.

This method helped streamline the app's functionality while ensuring it integrates seamlessly with current practices and meets compliance requirements.

4.4. Discussion/ Analysis of Results

4.4.1 Interview Analysis

To gain insights into the requirements for the Physio Connect App, interviews were conducted with physiotherapists and athletes. Physiotherapists shared their expectations, emphasizing the need for features that support remote monitoring, progress tracking, and exercise guidance. Athletes highlighted the importance of rehabilitation tools and how a digital solution could facilitate quicker recovery and injury prevention. These discussions provided critical information to shape the app's functionality, ensuring it meets the needs of both professionals and end-users.

4.4.2 Survey and Questionnaire Analysis

Question 01

This question was designed to collect email addresses of respondents for future communication. No analysis is provided as it only serves as a data collection mechanism.

Question 02

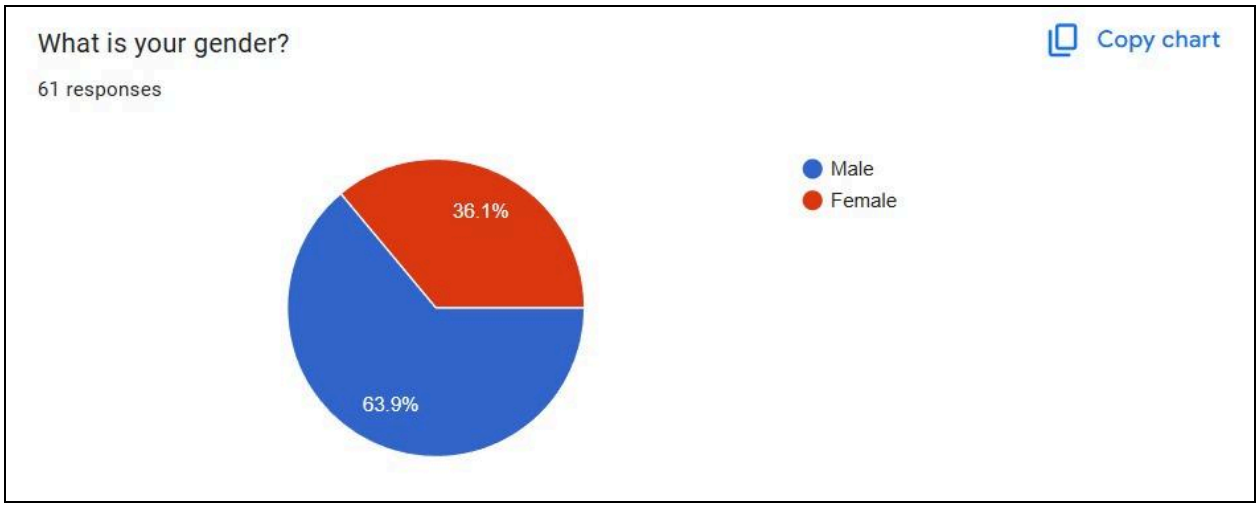


Figure 02: Survey Question-02

Analysis: The survey revealed that 63.9% of the respondents were male, reflecting the gender demographic of those interested in or aware of physiotherapy solutions.

Question 03

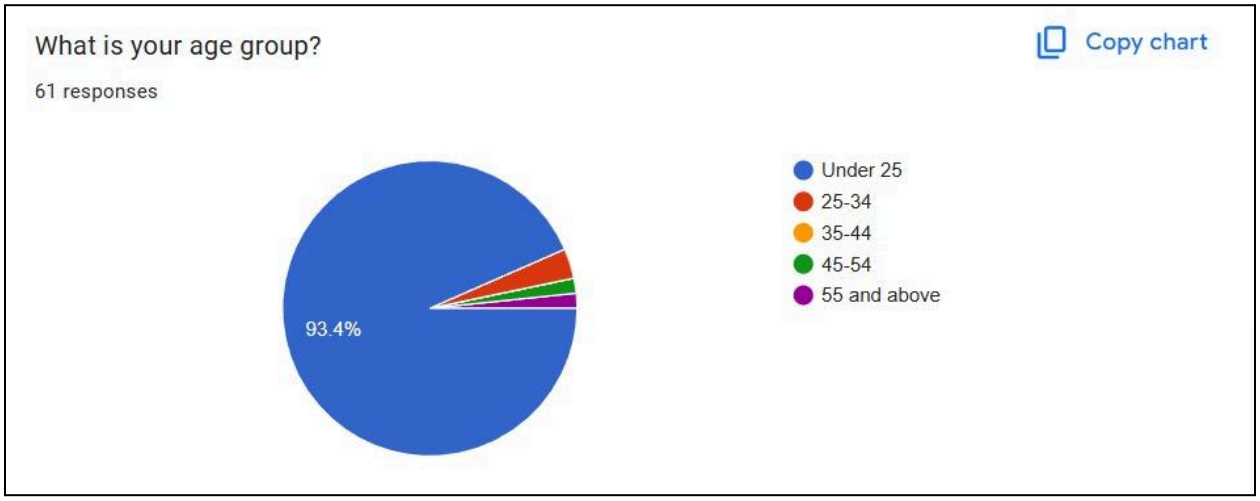


Figure 03: Survey Question-03

Analysis: The majority of respondents were under the age of 25, indicating that younger individuals are more inclined to adopt digital solutions like the Physio Connect App.

Question 04

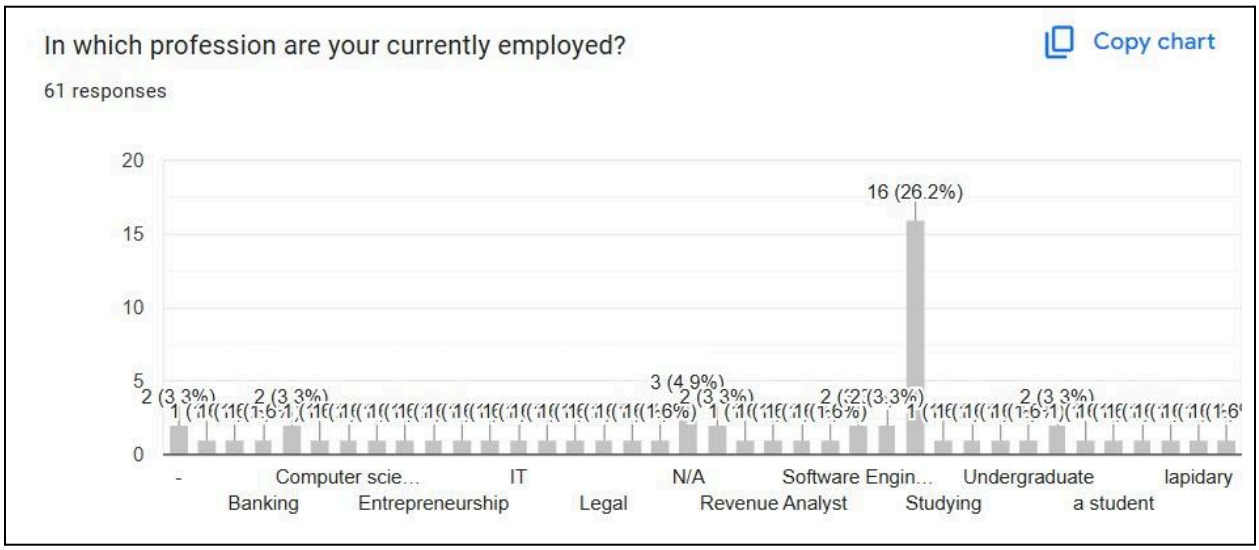


Figure 04: Survey Q4

Analysis: Most of the responses came from students studying at universities, schools, and colleges. This suggests a strong interest in the app among the student population, likely due to their active lifestyles and the prevalence of physical strain from studies and extracurricular activities.

Question 05

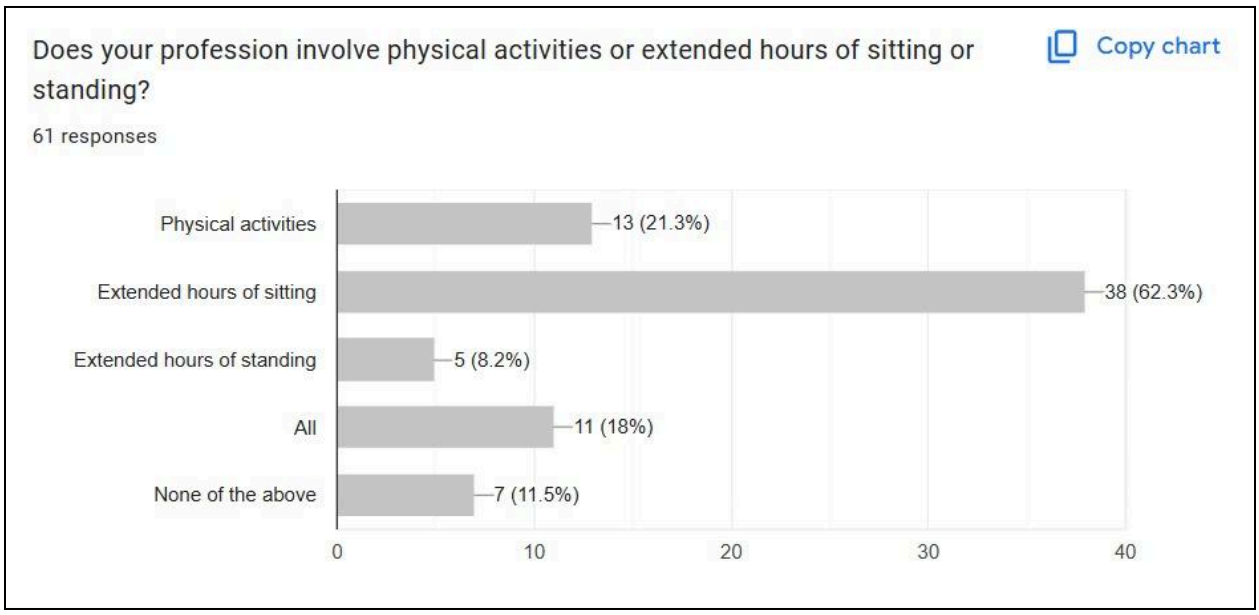


Figure 05: Survey Question-05

Analysis: The responses indicate that a significant number of survey participants have long sitting hours in their daily routines. This highlights a potential user base for the app, as prolonged sitting is a known contributor to musculoskeletal issues.

Question 06

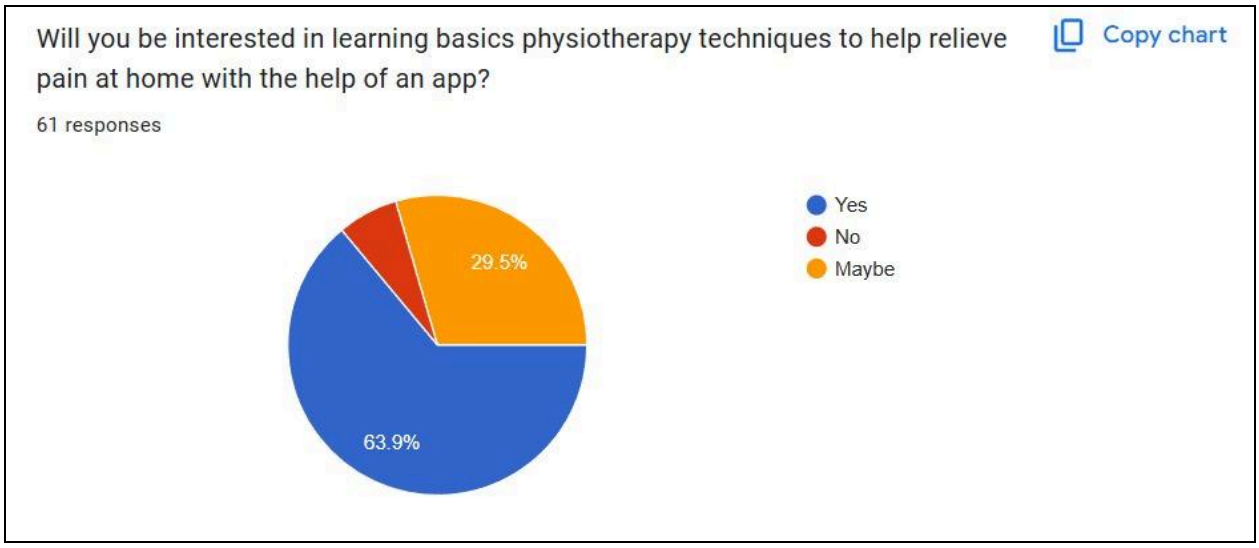


Figure 06: Survey Question-06

Analysis: Among those with long sitting hours, 13.1% reported sitting or standing for more than 8 hours daily, while 34.4% reported 4-6 hours. This data underscores the importance of preventive exercises and ergonomic advice, which the app aims to provide.

Question 07

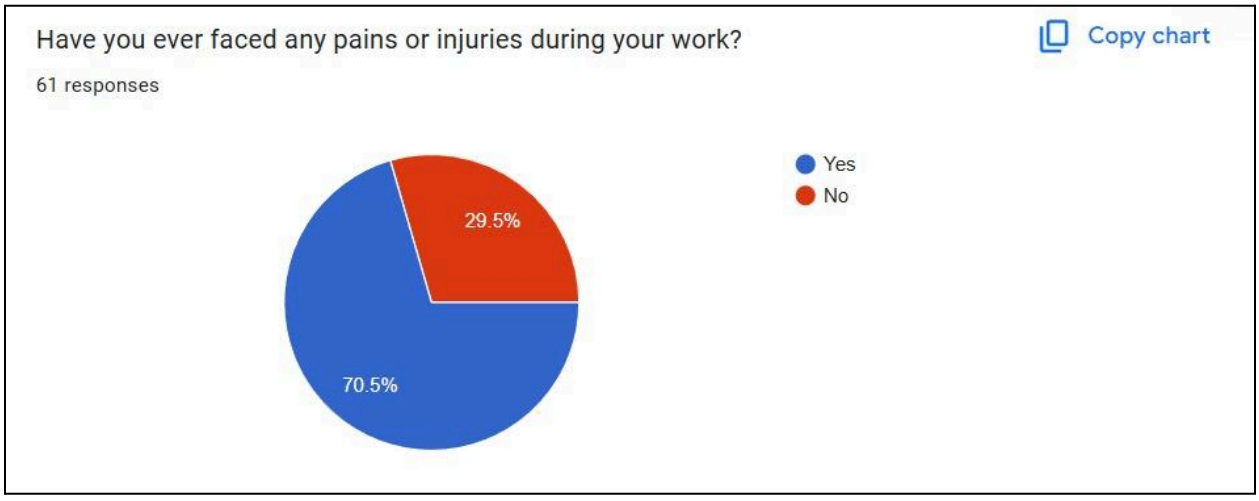


Figure 07: Survey Question-07

Analysis: An overwhelming 70.5% of respondents reported experiencing pain or injuries during work. This validates the need for the Physio Connect App to offer features focused on pain management and injury prevention.

Question 08

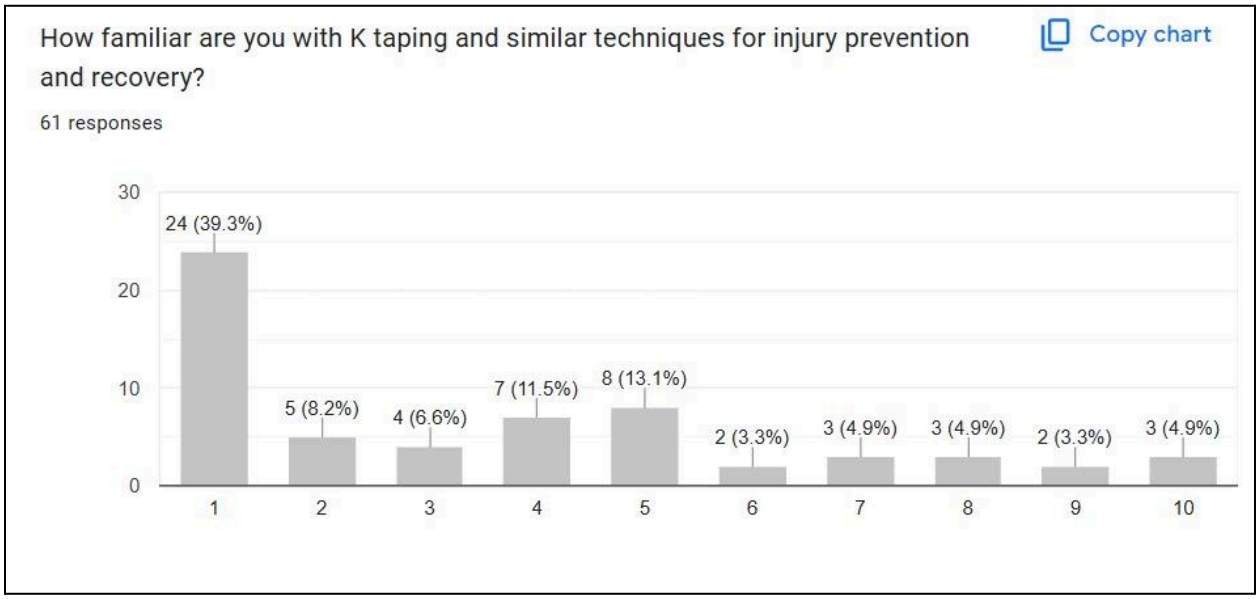


Figure 08: Survey Question-08

Analysis: A small percentage of respondents demonstrated familiarity with techniques like K-taping and injury prevention strategies. This suggests an opportunity to educate users on these methods through the app's content and features.

Question 09

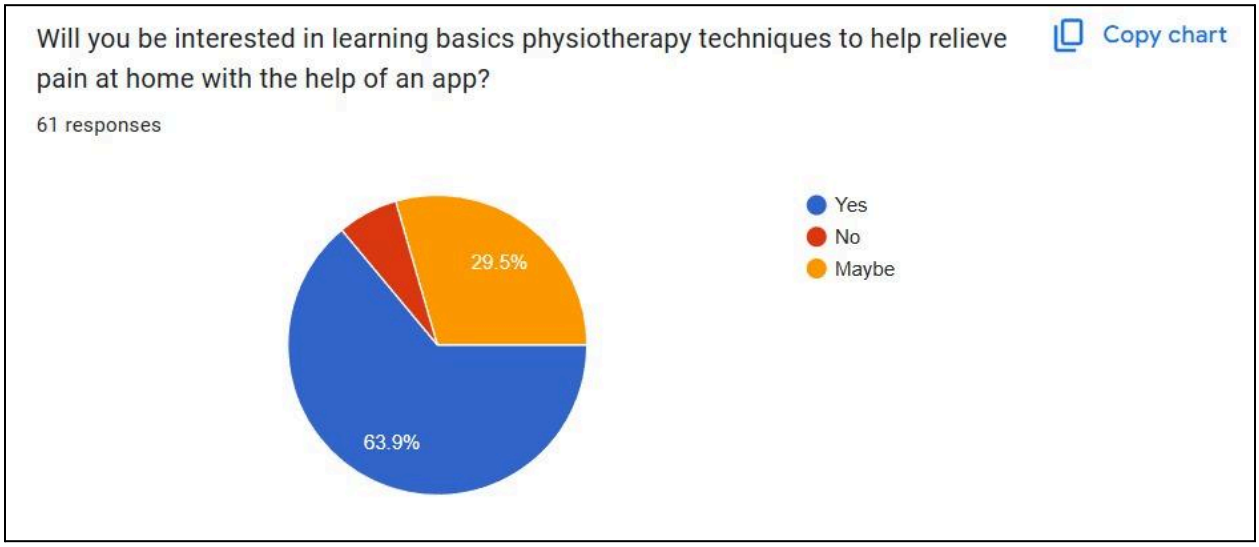
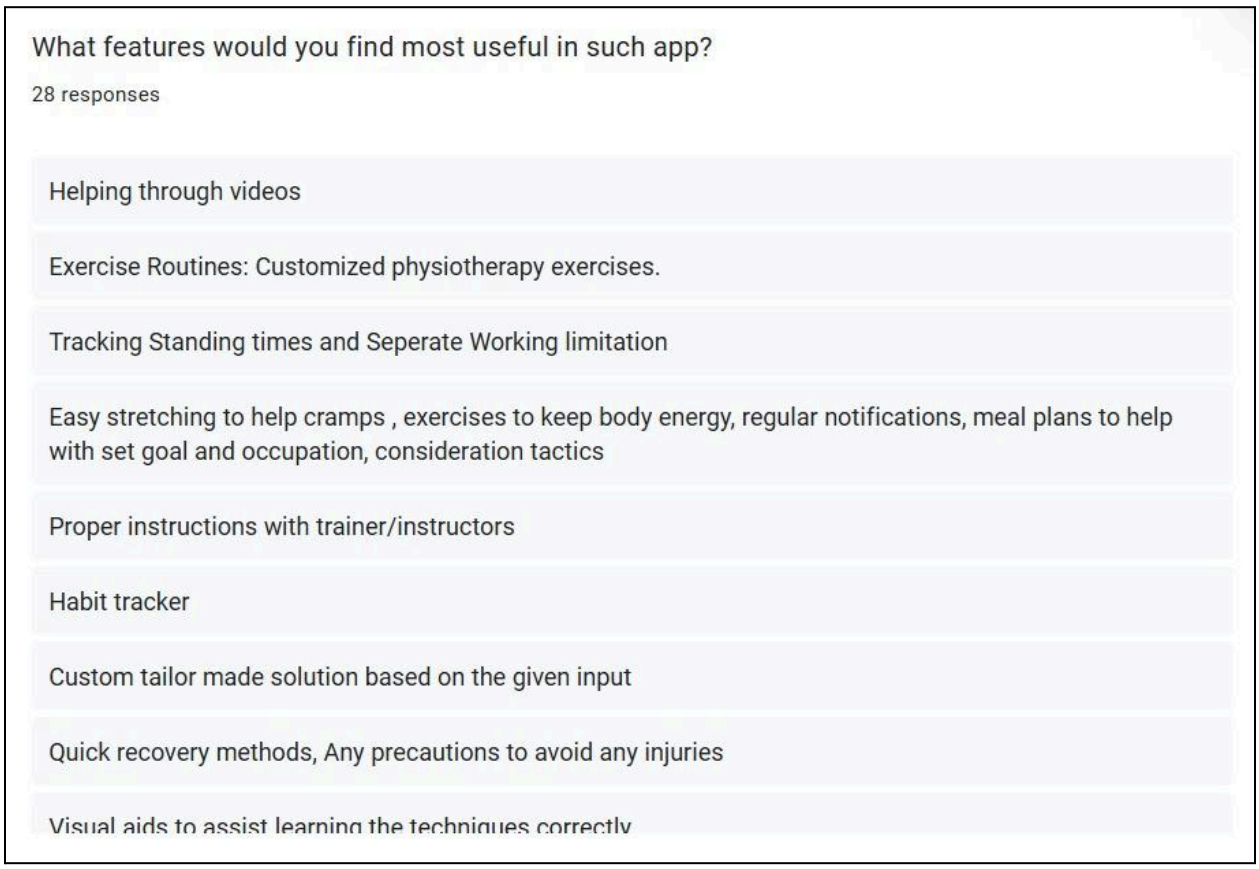


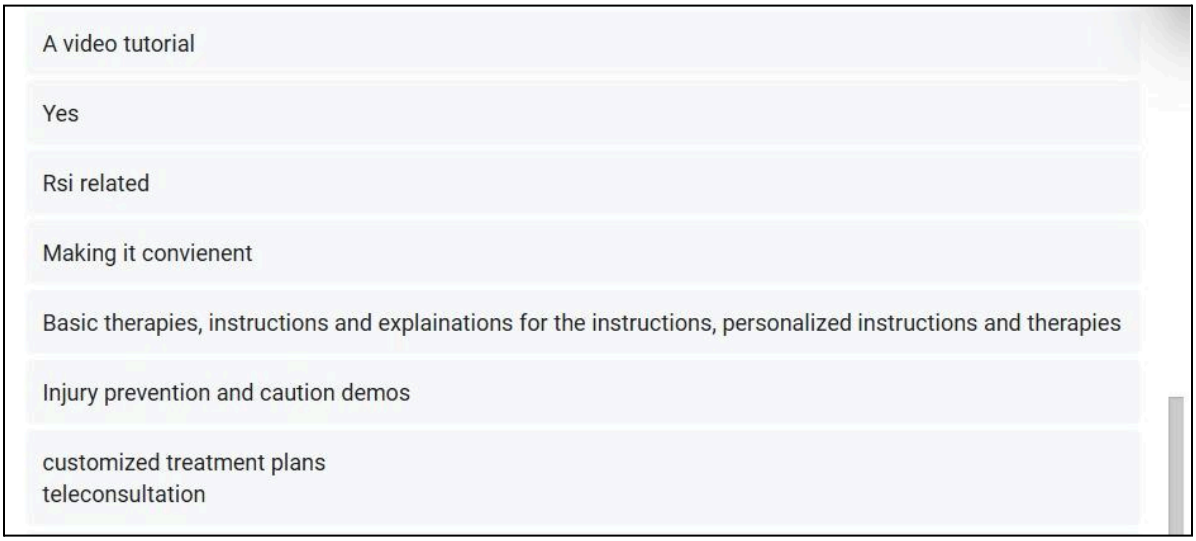
Figure 09: Survey Question-09

Analysis: Most respondents expressed a strong interest in learning physiotherapy techniques to alleviate pain and improve their quality of life at home. This highlights the need for easy-to-follow tutorials and guided exercises within the app.

Question 10



Figures 10 : Survey Question-10(A)



Figures 11 : Survey Question-10(B)

Analysis: Respondents provided various suggestions for features that could enhance the app, such as personalized exercise plans, injury recovery tracking, and virtual physiotherapy consultations. These inputs are being considered to prioritize features for development.

Question 11

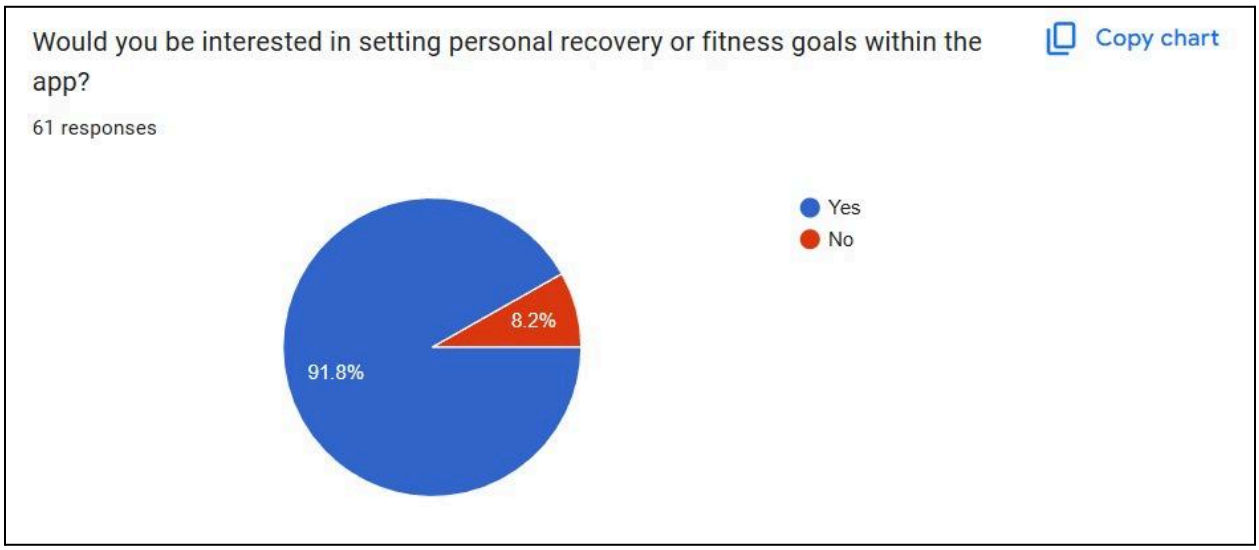


Figure 12: Survey Question-11

Analysis: The majority of respondents expressed a strong preference for having personal recovery or fitness goals integrated into the app. This feedback emphasizes the importance of goal-setting features that foster motivation and track progress.

4.5. Use Case Diagram

The following Use Case Diagram will provide the basic understanding of how users will interact with the PhysioConnect App.

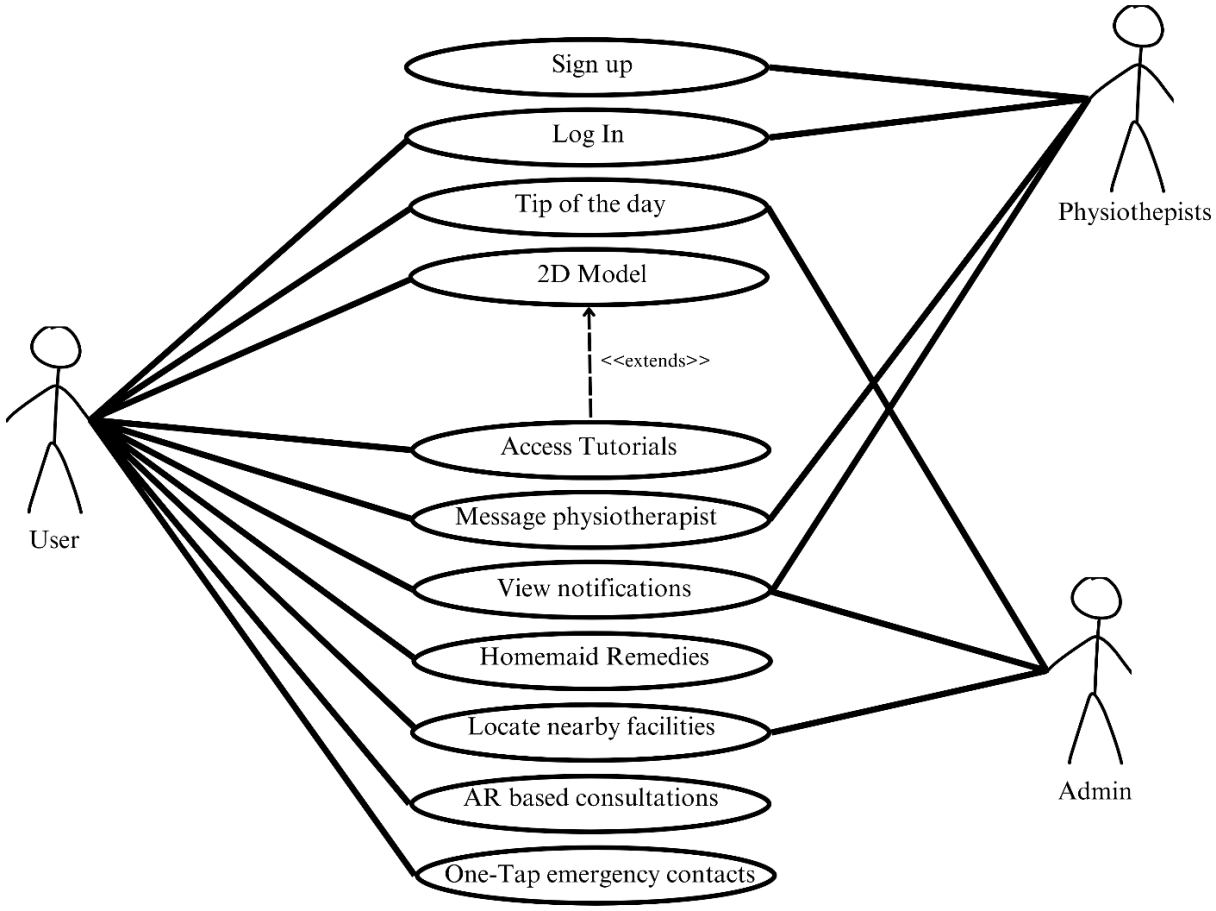


Figure 13: Use Case Diagrams

4.6. Use Case Descriptions

4.6.1 Sign Up

Use Case Name	Sign up	
Use Case ID	UC-001	
Description	Users and Physiotherapists can register themselves and create an account for themselves.	
Priority	High	
Primary Actor	User, Physiotherapist	
Supporting Actors	None	
Pre-Conditions	The application has to be installed by the user /physiotherapist.	
Trigger	The user/physiotherapist chooses the “Sign Up” option upon opening the app..	
Main flow	Actors	System
	1. The user/physiotherapist enters a username and a strong password to create an account.	2. The system verifies the details entered 3. It creates a new account for the users/physiotherapist.
Exception flow	Actors	System
	1. If the information entered are incorrect	2. The user or the physiotherapist is prompted by the system to make changes.

Alternate flow	Actors	System
	None	None
Exclusions	None	
Post Conditions	The users or the physiotherapists can login since they have an account that is registered.	

Table 03 : Use Case Description - Sign Up

4.6.2 Log In

Use Case Name	Log In	
Use Case ID	UC-002	
Description	The registered users/physiotherapists enter their login information to access the physio connect app.	
Priority	High	
Primary Actor	User, Physiotherapist	
Supporting Actors	None	
Pre-Conditions	The user/ physiotherapist should know their login credentials.	
Trigger	The user/physiotherapist chooses the ‘Login’ option.and enter their credentials.	
Main flow	Actors	System
	1. user/physiotherapist input their username and password.	2. The credentials are verified by the system. 3. The system provides

		access for the application if it is vaid.
Exception flow	Actors	System
	1. When the entered credentials aren't valid	2. The user/physiotherapist is prompted by the system to enter valid credentials.
Alternate flow	Actors	System
	None	None
Exclusions	None	
Post Conditions	The user/physiotherapist is logged in and can use the app features.	

Table 04 : Use Case Description - Login

4.6.3 Tip of the Day

Use Case Name	Tip of the Day
Use Case ID	UC-003
Description	This is where physiotherapy related tips are given to user
Priority	Medium
Primary Actor	User
Supporting Actors	Admin

Pre-Conditions	The user should have logged in to the app	
Trigger	Click on the 'Tip of the day' section.	
Main flow	Actors	System
	1. User clicks on the tip of the day option.	2. Fetches data from the database an 3. Displays it on the screen
Exception flow	Actors	System
	None	None
Alternate flow	Actors	System
	None	None
Exclusions	None	
Post Conditions	Users can access daily tips	

Table 05 : Use Case Description - Tip of the Day

4.6.4 2D Model

Use Case Name	2D Model
Use Case ID	UC-004
Description	This is where the user click on a specific area on the 2d model
Priority	High

Primary Actor	User	
Supporting Actors	None	
Pre-Conditions	The user must login to the application	
Trigger	Click on the area which is needed for treatment on the model	
Main flow	Actors	System
	1. The user clicks on the area he/she needs therapy	2. The system gives the required treatment plans and give tutorials
Exception flow	Actors	System
	None	None
Alternate flow	Actors	System
	None	None
Exclusions	None	
Post Conditions	The user gets the needed tutorial and recovery treatment plans for the injury.	

Table 06 : Use Case Description - 2D Model

4.6.5 Message Physiotherapists

Use Case Name	Message Physiotherapists	
Use Case ID	UC-005	
Description	Users can message physiotherapists for advice or for any clarification.	
Priority	Medium	
Primary Actor	User	
Supporting Actors	Physiotherapist, Admin	
Pre-Conditions	User and the physiotherapist should have logged in.	
Trigger	Click on the ‘message physiotherapist’ option.	
Main flow	Actors	System
	1. The user selects a specific physiotherapist. 2. The user inputs the message to be sent	3. The system sends the message to the selected physiotherapist
Exception flow	Actors	System
	1. If the physiotherapist is not available.	2. The system will informs the user about it.
Alternate flow	Actors	System
	None	None
Exclusions	None	

Post Conditions	Gets a response from the physiotherapist for the question.

Table 07 : Use Case Description - Message Physiotherapists

4.6.6 Access Tutorials

Use Case Name	Access Tutorials	
Use Case ID	UC-006	
Description	Once the user clicks on the 2D model at a specific area for treatment it gives tutorials related to it.	
Priority	High	
Primary Actor	User	
Supporting Actors	None	
Pre-Conditions	Have logged in to the app	
Trigger	Click on the area in the 2D model	
Main flow	Actors	System
	1. The user click on the 2D model on the area he/she is having pain	2. The system gives out tutorial and other treatment plans for the pain recovery
Exception flow	Actors	System
	None	None

Alternate flow	Actors	System
	None	None
Exclusions	None	
Post Conditions	User receives tutorials related to his/her injury for recovery	

Table 08 : Use Case Description -Access Tutorials

4.6.7 View Notification

Use Case Name	View Notification	
Use Case ID	UC-007	
Description	The user/ physiotherapist receives notification from the software for related activities.	
Priority	High	
Primary Actor	User, Physiotherapist	
Supporting Actors	Admin	
Pre-Conditions	Must have enabled app notifications.	
Trigger	The user/physiotherapists select the notifications	
Main flow	Actors	System
	1. User receive notification from the system	2. System extract them from the database

		3. Display them to the users
Exception flow	Actors	System
	None	None
Alternate flow	Actors	System
	none	None
Exclusions	None	
Post Conditions	User or Physiotherapist are update with schedule of consultation sessions	

Table 09 : Use Case Description - View Notification

4.6.8 Home Remedies

Use Case Name	Home Remedies
Use Case ID	UC-008
Description	This shows the user the remedies that he/she can take at an injury at home.
Priority	High
Primary Actor	User
Supporting Actors	None
Pre-Conditions	Must input the data about the injury to the system

Trigger	Select the home remedy option	
Main flow	Actors	System
	1. User select home remedies option	2. The system gives out some remedies using ai and other references
Exception flow	Actors	System
	1. If the injury cannot be recovered using home remedies	2. The system will update the user
Alternate flow	Actors	System
	None	None
Exclusions	None	
Post Conditions	The remedies are provided to the user.	

Table 10 : Use Case Description - Home Remedies

4.6.9 Nearby Facility Locator

Use Case Name	Nearby Facility Locator
Use Case ID	UC-009
Description	This allows users to find nearby health facility centers at any injury or treatment.
Priority	High

Primary Actor	User	
Supporting Actors	None	
Pre-Conditions	Must log in to the app.	
Trigger	Find nearby	
Main flow	Actors	System
	1. The user selects the nearby facility locator option	2. The system will access the user's location 3. Then it will give the nearby facilities in his/her area.
Exception flow	Actors	System
	1. If there are no any facility center nearby	2. The system will notify the user
Alternate flow	Actors	System
	None	None
Exclusions	None	
Post Conditions	The user will view the nearby facility.	

Table 11 : Use Case Description - Nearby Facility Locator

4.6.10 AR Based Consultation

Use Case Name	AR Based Consultation	
Use Case ID	UC–010	
Description	Allows the user to consult physiotherapists through augmented reality.	
Priority	High	
Primary Actor	User	
Supporting Actors	Physiotherapist	
Pre-Conditions	Both users and physiotherapists must have AR compatible devices.	
Trigger	Select the ‘AR Based Consultation’ Option	
Main flow	Actors	System
	1. The user select AR consultation option	2. The system connects a physiotherapist
Exception flow	Actors	System
	1. If the service in not available	2. The system will notify the user about the unavailability
Alternate flow	Actors	System
	None	
Exclusions	None	

Post Conditions	The user will get consulted via AR

Table 12 : Use Case Description - AR Based Consultation

4.6.11 One-tap Emergency Contact

Use Case Name	One-tap Emergency Contact	
Use Case ID	UC-011	
Description	Allows users to quickly contact any emergency service or any emergency contact saved.	
Priority	High	
Primary Actor	User	
Supporting Actors	None	
Pre-Conditions	Must have at least one number saved in the emergency contact list.	
Trigger	The user select the One-Tap Emergency Contact button	
Main flow	Actors	System
	1. The user select the emergency contact	2. The system dials the emergency contact
Exception flow	Actors	System
	1. If no number is saved in the emergency contact list	2. The system asks the user to enter a contact to the list.

Alternate flow	Actors	System
	None	None
Exclusions	None	
Post Conditions	The user’s emergency contact is contacted.	

Table 13 : Use Case Description - One-tap Emergency Contact

4.7 Functional Requirements (with prioritization)

The following table will discuss the function requirements of this project, as well the priority level of each requirement.

Critical - The Requirements that are critically needed for successful completion

Desirable - The Requirements that can add value, but are not required immediately

Luxury - The Requirement that would add luxury to the system

Requirement List		Priority Level	Description
FR1	User Registration/ Login	Critical	Enables users to securely create accounts and log in, ensuring the safe storage of personal and health-related data.
FR	Customized Exercise Plan	Critical	Generates personalized exercise routines tailored to users' injuries and recovery goals, ensuring effective rehabilitation.
FR3	Progress Tracking	Critical	Offers detailed insights into users' rehabilitation journeys, helping them monitor improvements and stay

			motivated.
FR4	Educational Resources	Critical	Provides a library of articles, tutorials, and guides on injury management, recovery techniques, and preventive measures to enhance user knowledge.
FR5	2D Body Model	Critical	An interactive tool that helps users identify injury locations and receive region-specific advice and exercises for targeted recovery.
FR6	Book Appointments	Desirable	Allows users to schedule appointments with physiotherapists for in-person consultations and therapy sessions.
FR7	Video Consultations	Desirable	Facilitates remote assessments and treatment plans through video calls with physiotherapists, enabling care from the comfort of home.
FR8	Electronic Personal Health Record	Desirable	Centralizes users' health data, including injury history, recovery progress, and consultation notes, for easy access and management.
FR9	Messaging System	Desirable	Provides direct communication between users and physiotherapists, ensuring ongoing support and quick resolution of queries.
FR10	Nearby Facilities Locator	Desirable	Helps users find the nearest physiotherapy centers or medical facilities using

			geo-location technology for added convenience.
FR11	Push Notification	Desirable	Sends reminders for appointments, exercise routines, and daily health tips, encouraging adherence and consistency.
FR12	Emergency Contact	Desirable	Offers one-tap access to emergency contacts, ensuring immediate assistance in critical situations.
FR13	AI Chatbot for Advice	Luxury	Provides real-time physiotherapy advice and tips, offering users instant support through an intelligent virtual assistant.
FR14	Payment Integration	Luxury	Allows users to pay for consultations, packages, or services directly within the app, streamlining the payment process.
FR15	AR Implementation	Luxury	Uses AR technology to guide users through exercises with real-time posture correction and movement feedback for improved accuracy.
FR16	Gamification	Luxury	Introduces challenges, point systems, and rewards to increase user engagement and make rehabilitation more interactive and enjoyable.

Table 14 : Functional Requirements

4.8 Non-Functional Requirements

The following table will discuss the non functional requirements of this project, as well the priority level of each requirement.

Critical - The Requirements that are critically needed for successful completion

Desirable - The Requirements that can add value, but are not required immediately

Luxury - The Requirement that would add luxury to the system

Requirement list		Priority Level	Description
NFR1	Response Time	Critical	This is where the app should respond to user actions in a short period of time to improve user experience.
NFR2	Scalability	Critical	This is where the app must support increasing user volume without performance issues.
NFR3	Data Encryption	Critical	During Storage user data must be encrypted.
NFR4	Authentication	Critical	This uses a safe authentication system to protect user accounts.
NFR5	Availability	Critical	This ensures 99.9% uptime with minimal disruptions during maintenance.
NFR6	Accessibility	Desirable	This Complies with standards to ensure usability for

			all,including disabled users.
NFR7	Backup & Recovery	Desirable	This performs daily backups and ensures recovery in case of system failure.
NFR8	Cross-Platform Compatibility	Desirable	This provides a consistent experience across a range of operating systems.
NFR9	User Training Resources	Luxury	This provides materials to help users utilize the app's features fully.
NFR10	Performance Monitoring	Luxury	This Implements monitoring tools to supervise user interaction and system performance.

Table 15 : Non-Functional Requirements

4.9. Chapter Summary

A thorough evaluation of system requirements is critical for the success and seamless execution of any project. Defining these requirements with precision lays a strong foundation for achieving project objectives. This chapter delves into key aspects of our project, beginning with an exploration of stakeholders, illustrated through an onion model, along with detailed descriptions and analyses.

It further highlights the elicitation techniques adopted to gather and understand stakeholder requirements effectively. Functional and non-functional requirements are outlined, categorized

by their respective priority levels to ensure a clear understanding of project needs. The findings from the survey and questionnaire are analyzed to provide valuable insights into user preferences and expectations.

Additionally, the chapter incorporates use case diagrams and detailed descriptions, presenting the core concepts of the project and demonstrating how the system aligns with user goals. These elements collectively contribute to a comprehensive understanding of the system requirements and the framework for developing a successful solution.

Chapter 5: Social, Legal, Ethical and Professional Issues

5.1. Chapter Overview

This chapter delves into the broader implications of our project, emphasizing its integration with societal, legal, ethical, and professional dimensions. It highlights the responsibilities of administrators and stakeholders to uphold professional standards, address potential social impacts, and ensure compliance with legal and ethical frameworks. By focusing on these critical aspects, the chapter underscores the importance of building trust, safeguarding individual rights, and ensuring the long-term sustainability and success of the project.

5.2 Dataset Ethical Clearance

The datasets utilized in this project were sourced from Kaggle (www.kaggle.com), a well-known platform for open datasets. These datasets are made available under the Public Domain license granted by The Creative Commons License Organization, allowing unrestricted use for research purposes.

Patient-Related Dataset:

> This dataset provides insights into patient physical activity tracking and chronic pain management. (https://www.kaggle.com/search?q=Health+and+Lifestyle+Data)

Exercise and Rehabilitation Datasets:

> A comprehensive collection of health-related information, useful for analyzing patterns in patient recovery and overall health trends.
> (https://www.kaggle.com/search?q=Human+Activity+Recognition)

5.3 SLEP Issues and Mitigation

Developing the Physio Connect app required careful consideration of social, legal, ethical, and professional (SLEP) issues to ensure compliance with standards and guidelines while meeting user needs effectively. Below is an analysis of how these issues were addressed in alignment with the BCS Code of Conduct.

5.3.1 Social Issues

The Physio Connect app aims to expand access to physiotherapy services by offering virtual consultations and rehabilitation tools, especially in underserved or rural areas. However, challenges remain:

- **Digital Divide:** The app relies on smartphones and stable internet connectivity, which may disadvantage users in low-income areas or regions with inadequate digital infrastructure.
- **Technological Barriers:** Older adults or individuals unfamiliar with digital tools may struggle to use the app effectively.

Mitigation Strategies:

While the project primarily targets metropolitan and semi-urban areas with higher technology penetration, efforts have been made to simplify the app's interface to support users with limited technical expertise. Future phases will explore lightweight features and offline modes to bridge accessibility gaps in rural and underserved communities.

5.3.2 Legal Issues

Providing quality physiotherapy through the app requires adherence to complex healthcare regulations and legal frameworks.

- **Data Protection and Privacy:** Compliance with laws like GDPR and HIPAA is critical to ensure users' sensitive health data is safeguarded.

- **Professional Licensing:** Ensuring physiotherapists on the platform are certified and licensed according to regional laws.
- **International Consultations:** Addressing varying healthcare regulations and liability risks in cross-border virtual sessions.

Mitigation Strategies:

- Implemented robust encryption and privacy protocols to protect user data.
- Developed informed consent mechanisms for all consultations and data-sharing practices.
- Partnered with legal advisors to navigate licensing requirements and healthcare laws across different jurisdictions.

5.3.3 Ethical Issues

The Physio Connect app handles sensitive health data to deliver personalized care, raising ethical concerns around privacy, data usage, and fairness.

- **Data Usage:** Users must trust that their data will not be exploited or shared without explicit consent.
- **AI Bias:** Ensuring AI-driven recommendations remain unbiased and equitable to avoid preferential treatment that could disadvantage certain users.

Mitigation Strategies:

- Established clear policies on data collection and use, ensuring transparency and user control over their information.
- Conducted rigorous testing of AI algorithms to identify and address potential biases, prioritizing fairness and inclusivity in recommendations.

5.3.4 Professional Issues

Maintaining high standards of professionalism is critical to the app's credibility and effectiveness.

- **Accreditation and Licensing:** Ensuring all physiotherapists on the platform meet licensing and accreditation requirements in their respective regions.
- **Managing Expectations:** Delivering effective therapy in a virtual setting requires careful management of user expectations while ensuring services stay within the practitioners' scope of expertise.
- **Accountability:** Remote consultations pose challenges in maintaining accountability for therapy outcomes.

Mitigation Strategies:

- Introduced a thorough vetting and verification process for physiotherapists, including periodic updates to credentials.
- Established professional guidelines for remote therapy and integrated feedback loops to ensure accountability.
- Provided ongoing training for practitioners to align their services with professional and ethical standards in virtual environments.

5.4 Chapter Summary

This chapter explored the social, legal, ethical, and professional (SLEP) challenges that our project may encounter during its development and implementation. These challenges are expected to become more evident as the project progresses. To address these concerns, our team identified and analyzed relevant SLEP issues, ensuring a proactive approach to mitigation. We outlined specific strategies to minimize potential negative impacts from external factors while safeguarding against any unintended harm the project itself may cause. By addressing these considerations, we aim to uphold the integrity and sustainability of our initiative

Chapter 6: System Architecture & Design

6.1. Chapter Overview

This chapter delves into the core principles of structuring and organizing the software system for the Physio Connect app. It provides a comprehensive analysis of the app's architecture and design, focusing on its components, interfaces, and interactions. The primary objective is to offer a clear and detailed overview of the system's design, encompassing its architecture, individual elements, interconnections, and data flow within the system. This structured approach ensures a cohesive and efficient foundation for the app's development and functionality.

6.2. System Architecture Design

Layered architecture is one of the most commonly adopted architectural patterns, known for its structured and modular approach. It organizes the system into horizontal layers, each grouping components with related functions. The architectural design of our Physio Connect app follows this model, dividing the system into three primary layers:

1. **Presentation Layer:** This topmost layer serves as the user interface, enabling seamless interaction between the users and the app. It includes visual elements and components that handle user inputs and display outputs.
2. **Domain Logic Layer:** Situated beneath the presentation layer, this layer represents the app's core functionalities and business logic. It processes user inputs, applies the necessary rules, and coordinates data exchange between the layers.
3. **Data Storage Layer:** The foundation of the architecture, this layer is responsible for managing and storing the app's data, including user information, session logs, and recovery plans. It ensures secure and efficient data retrieval and storage.

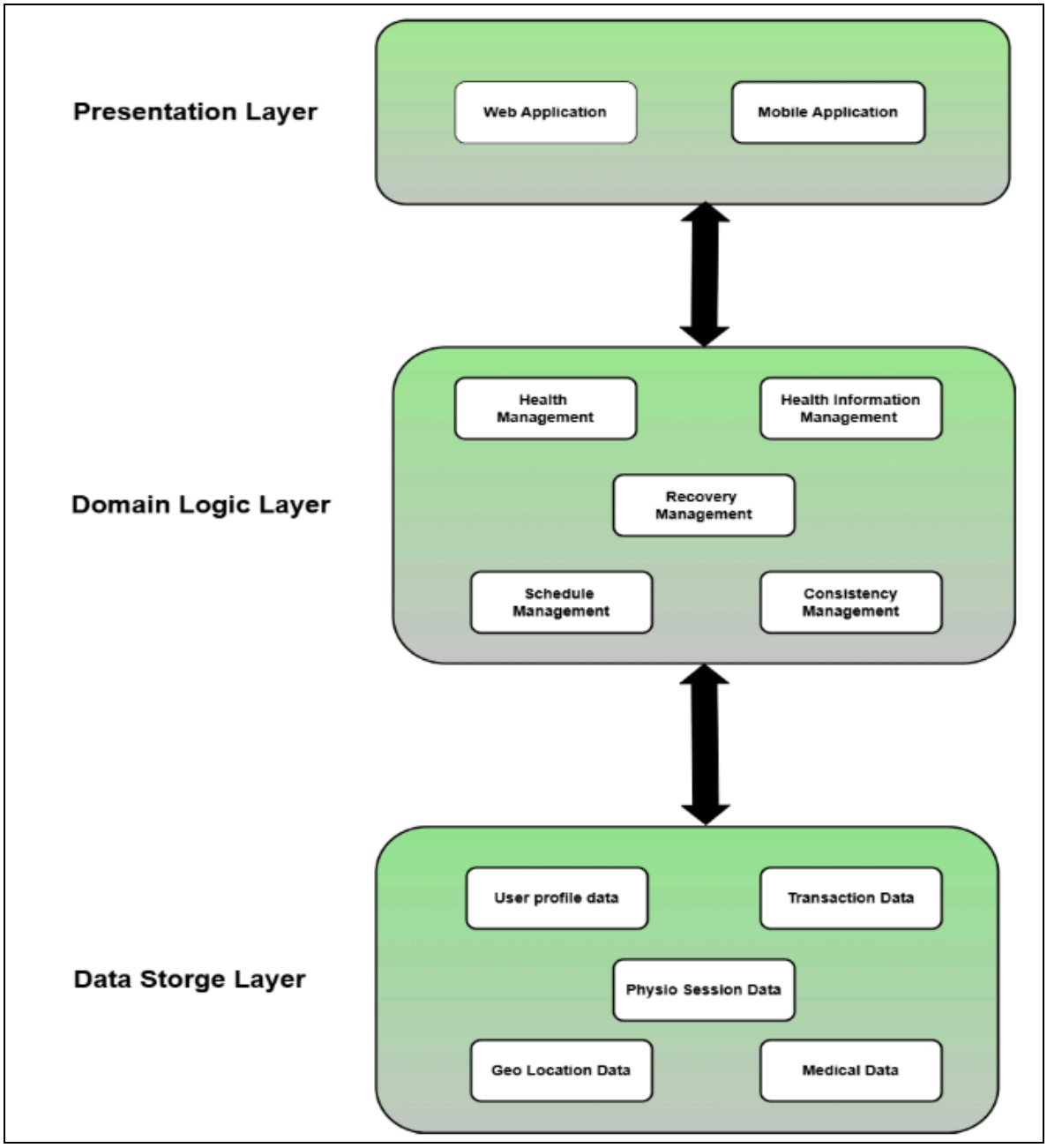


Figure 14: System Architecture Design

6.3. System Design

The system design outlines the structural and behavioral blueprint of the Physio Connect app, providing a clear representation of how different components interact and function cohesively.

6.3.1.Class Diagram/Data Flow Diagram (DFD)

Class Diagram Illustrates the static structure of the app by defining the classes, their attributes, methods, and the relationships between them. This helps to understand the key entities and their interactions within the system.

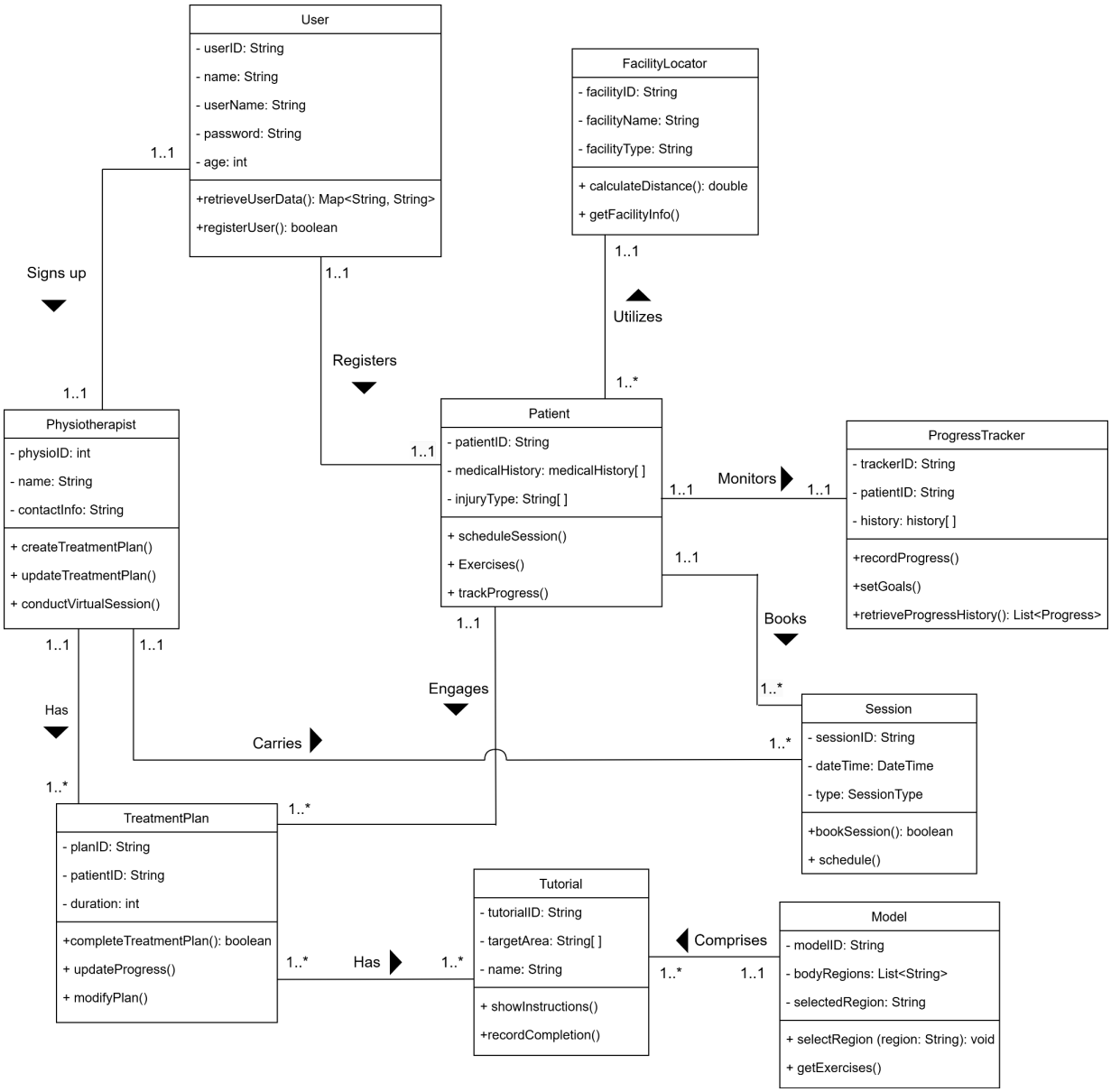


Figure 15: Class Diagram

6.3.2.Sequence Diagram

The sequence diagram details the dynamic interactions between system components in a step-by-step manner. It showcases the order of operations, highlighting how users and system entities communicate to execute functions like registering, booking consultations, and tracking progress. (STAR, 2024)

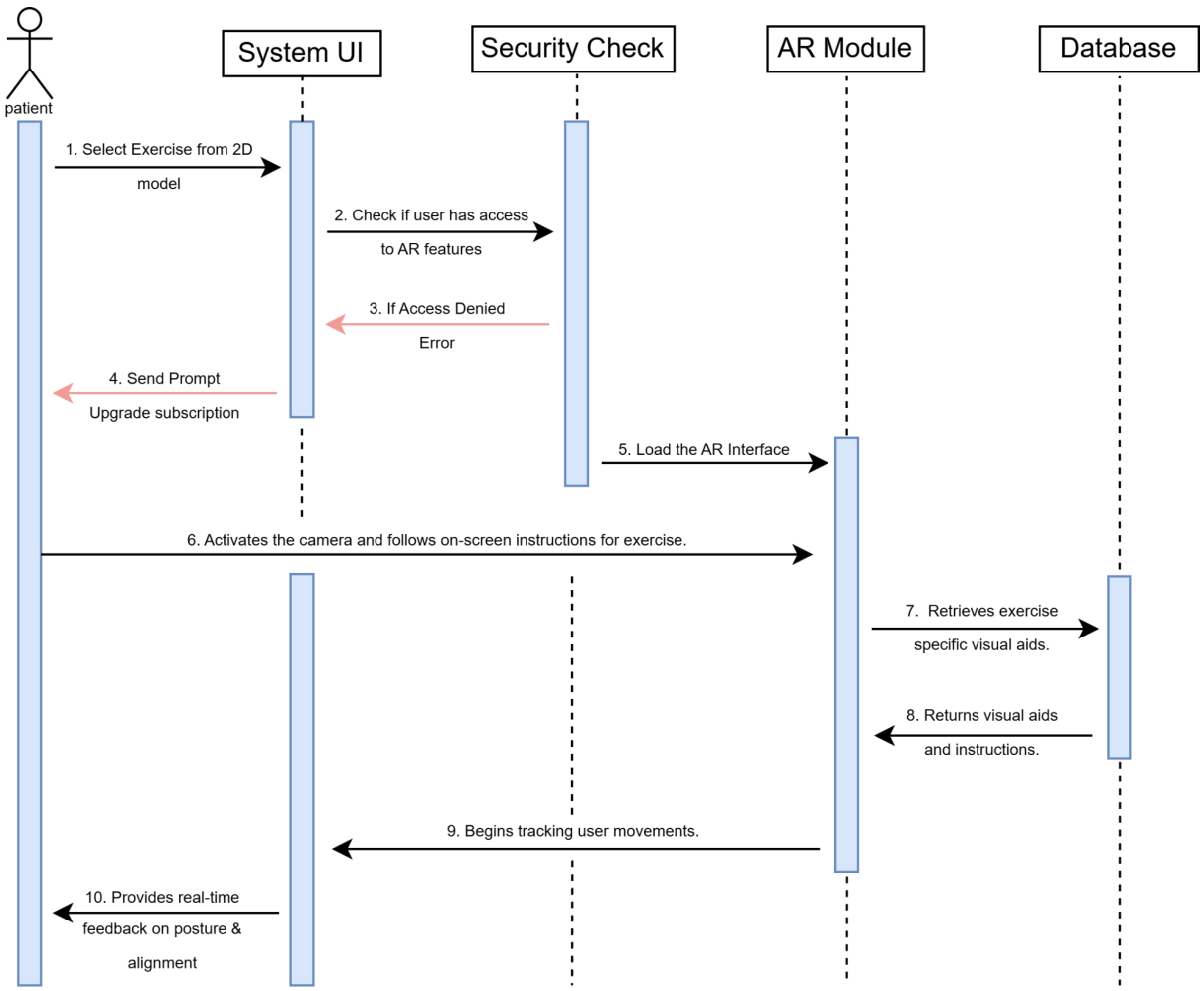


Figure 17: AR Guidance (Self Composed)

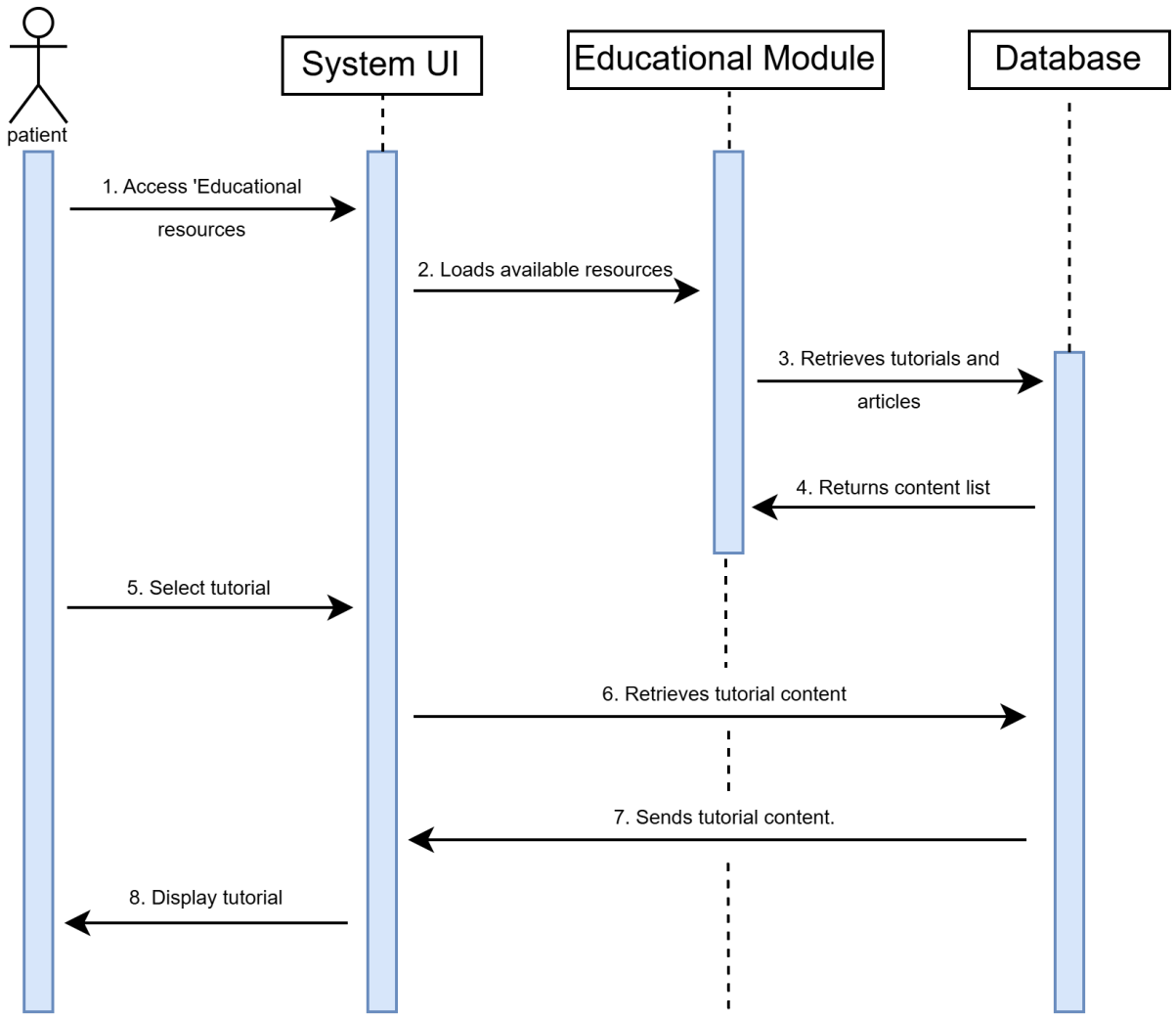


Figure 18: Educational Resources (Self Composed)

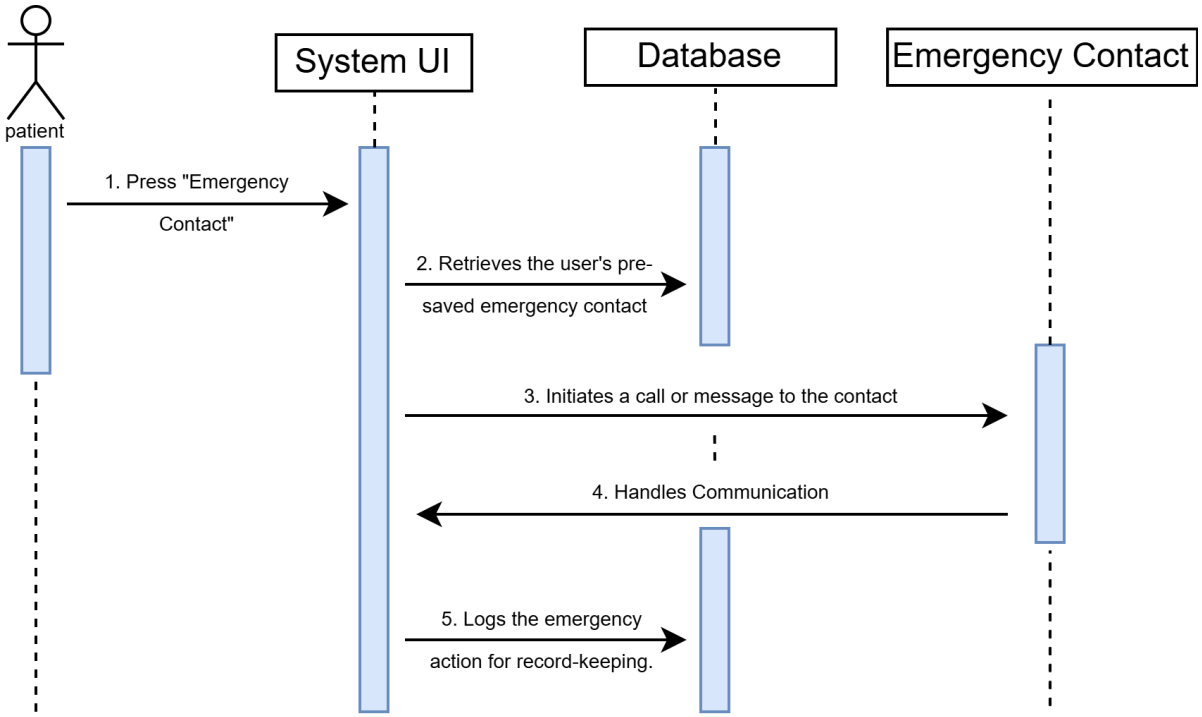


Figure 19: Emergency Contact (Self Composed)

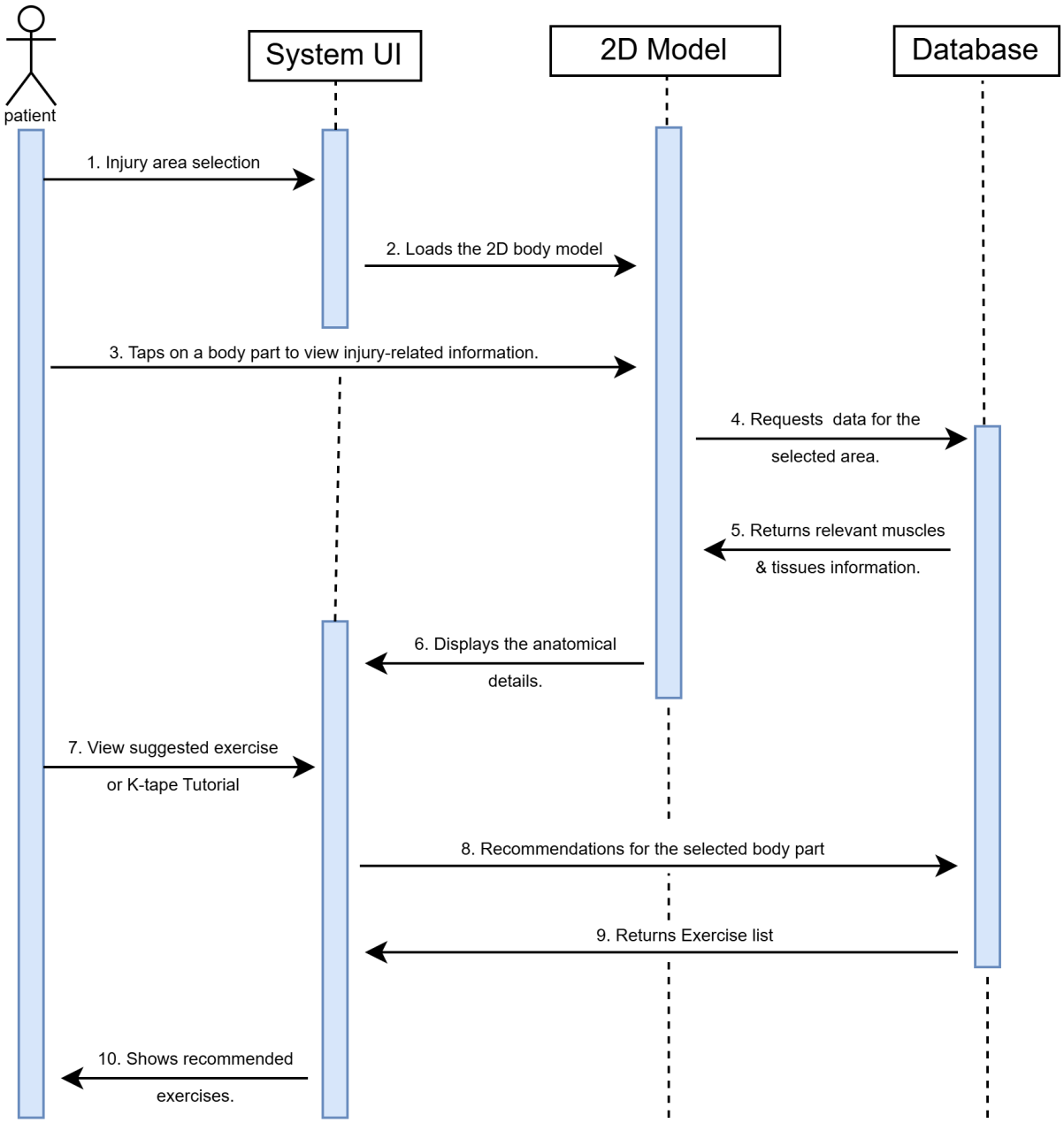


Figure 20: Injury Area Selection and 2D Model Interaction (Self Composed)

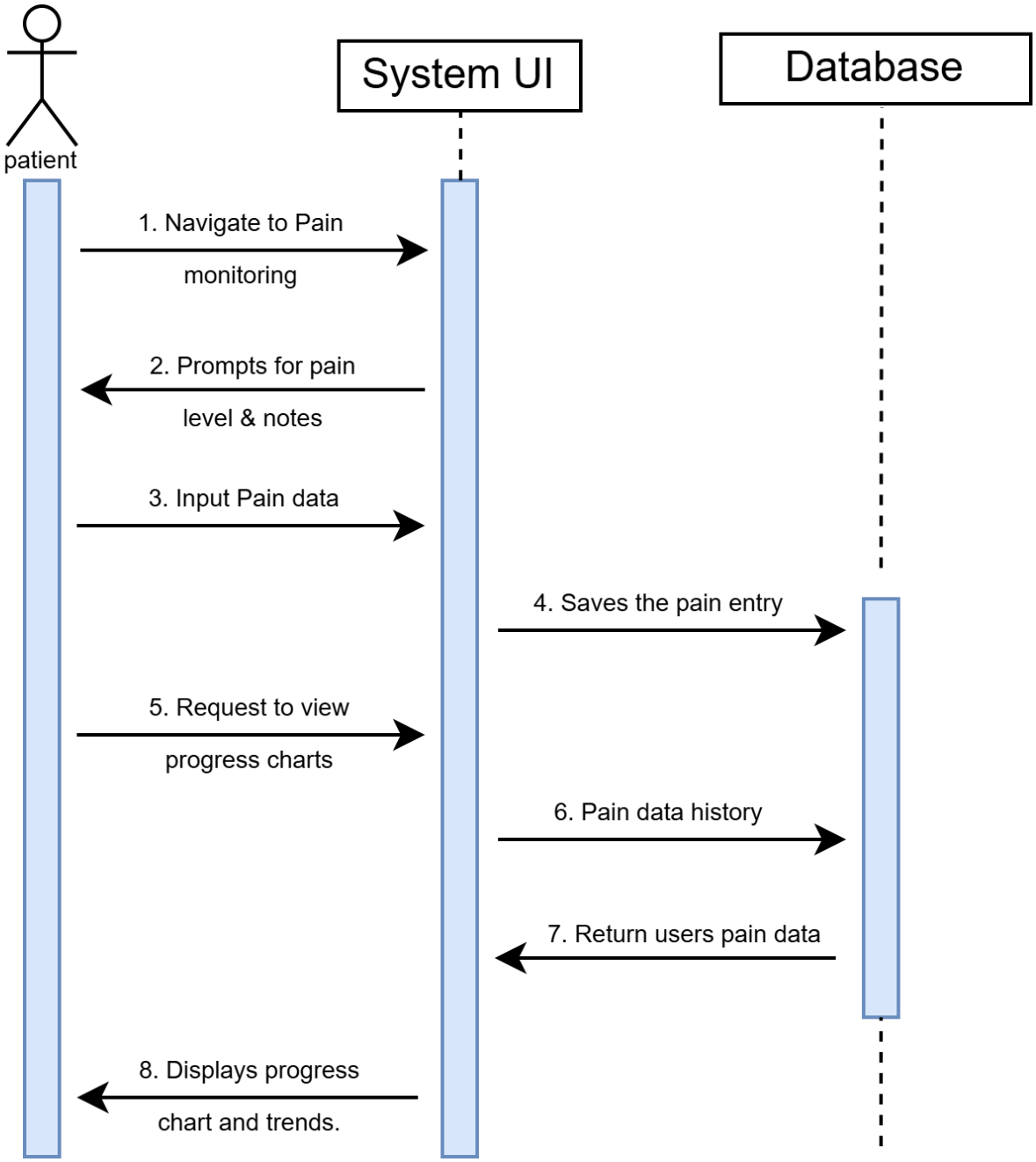

patient
System UI
Database
1. Navigate to Pain monitoring
2. Prompts for pain level & notes
3. Input Pain data
4. Saves the pain entry
5. Request to view progress charts
6. Pain data history
7. Return users pain data
8. Displays progress chart and trends.

Figure 21: Progress Tracking (Self Composed)

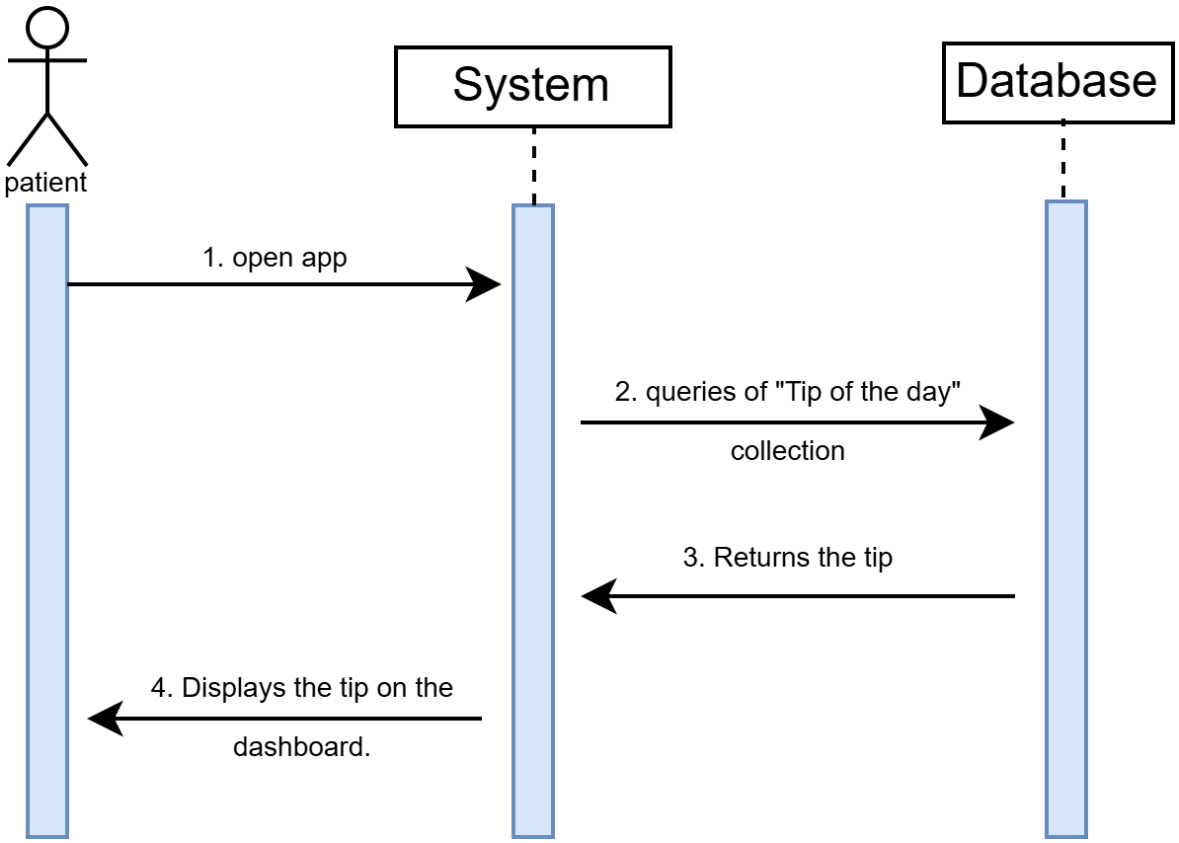


Figure 22: Tip of the Day (Self Composed)

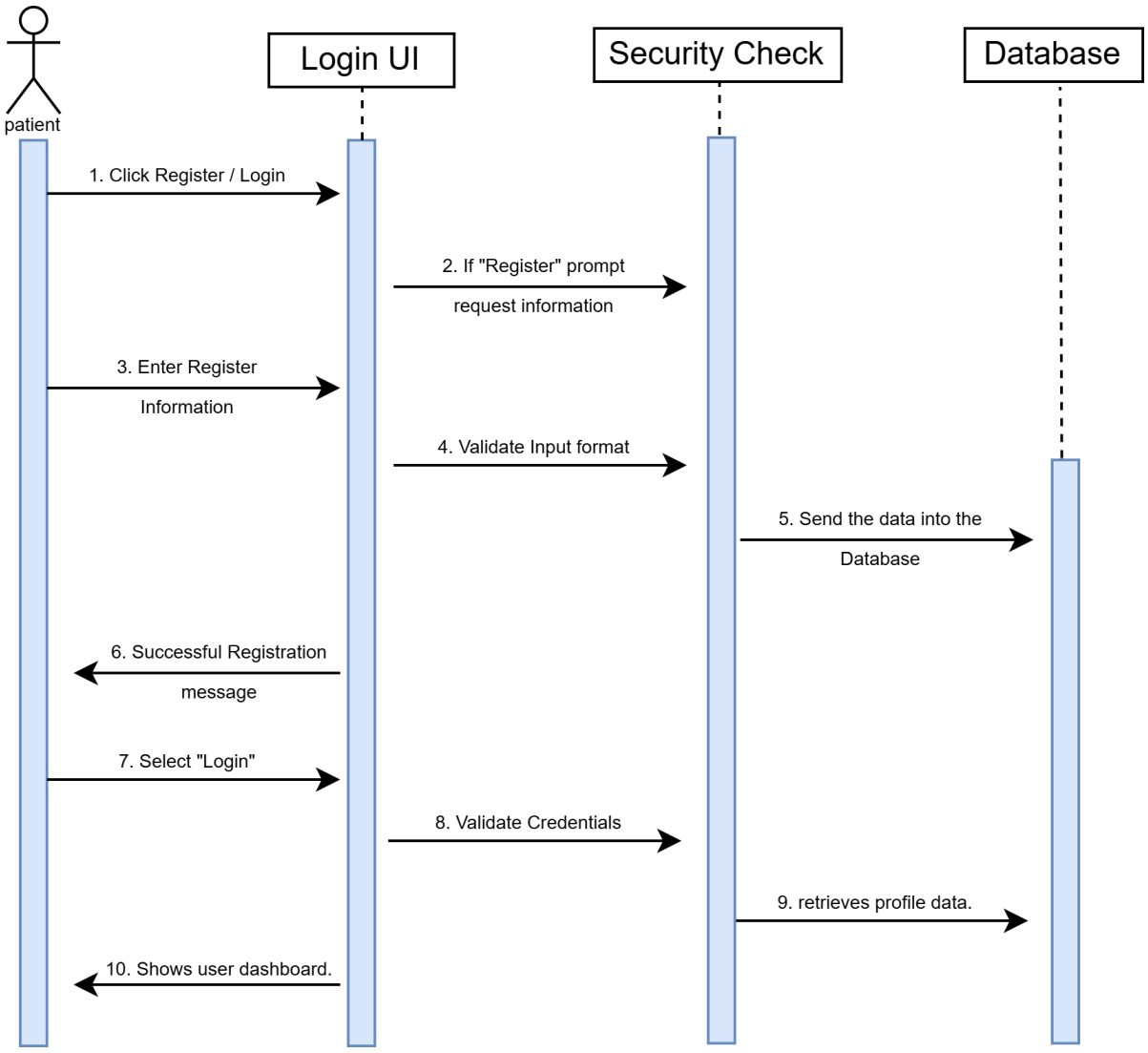


Figure 23: User Registration and Login (Self Composed)

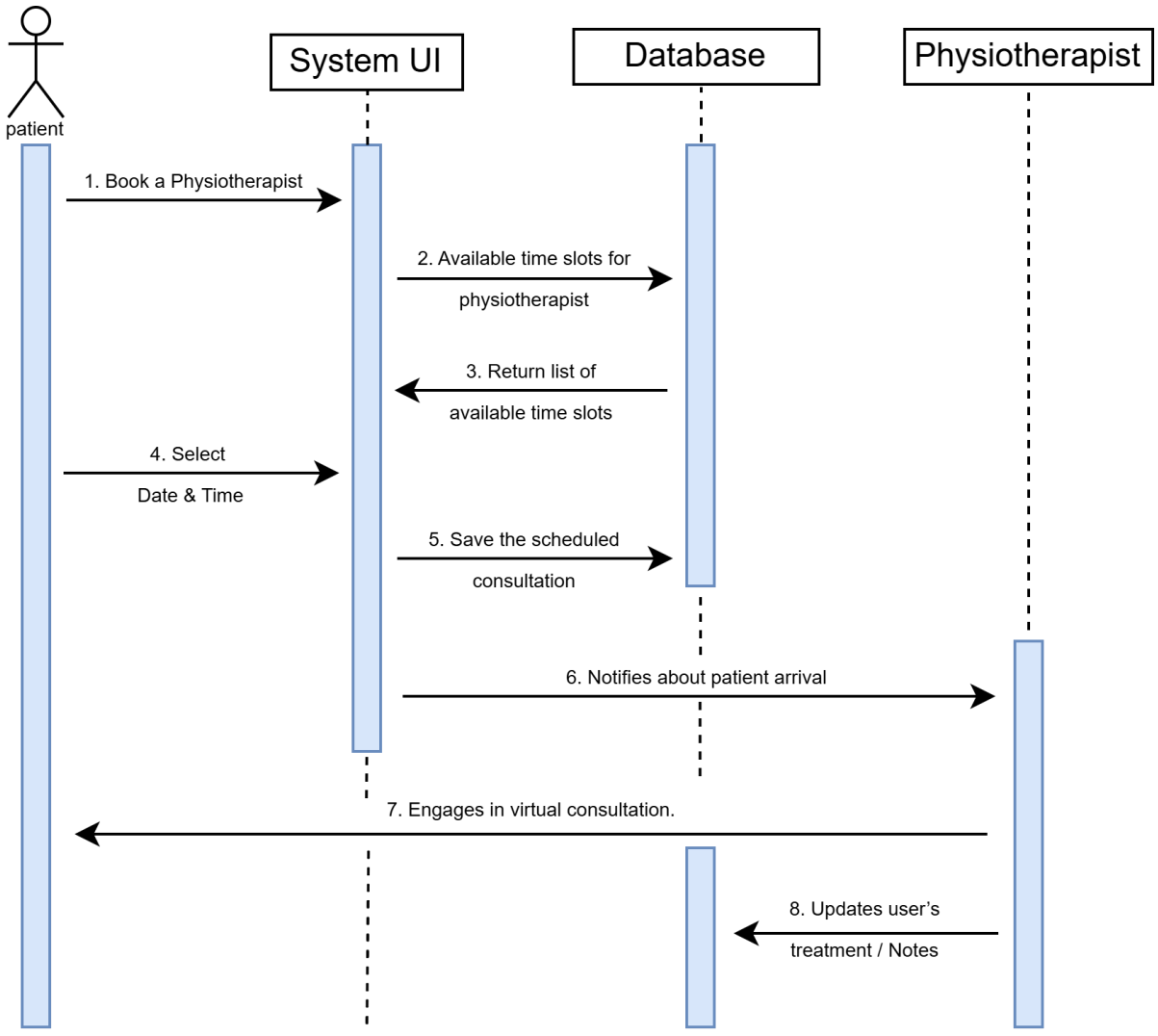
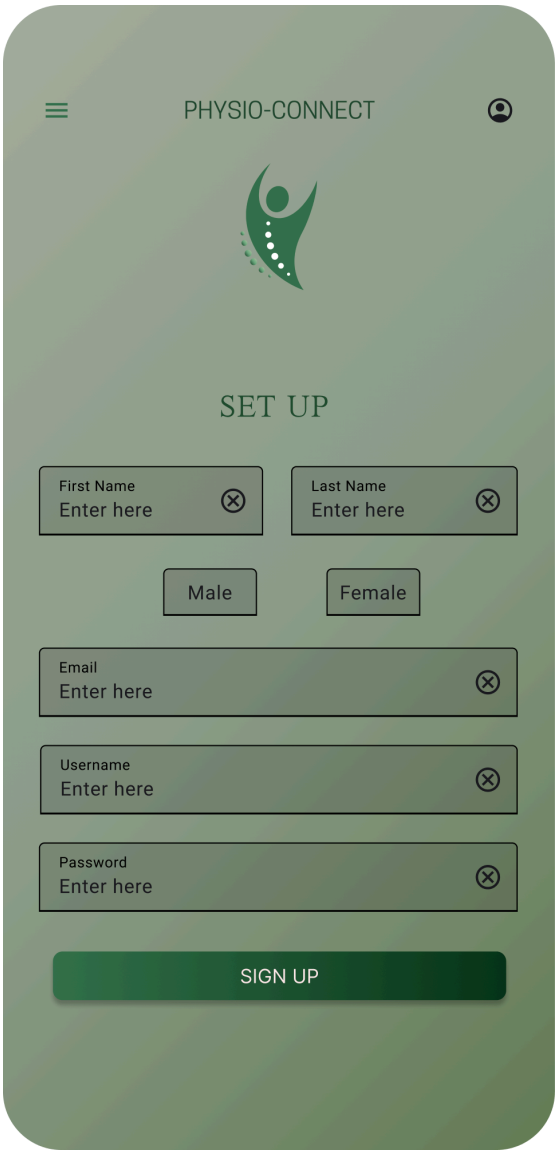


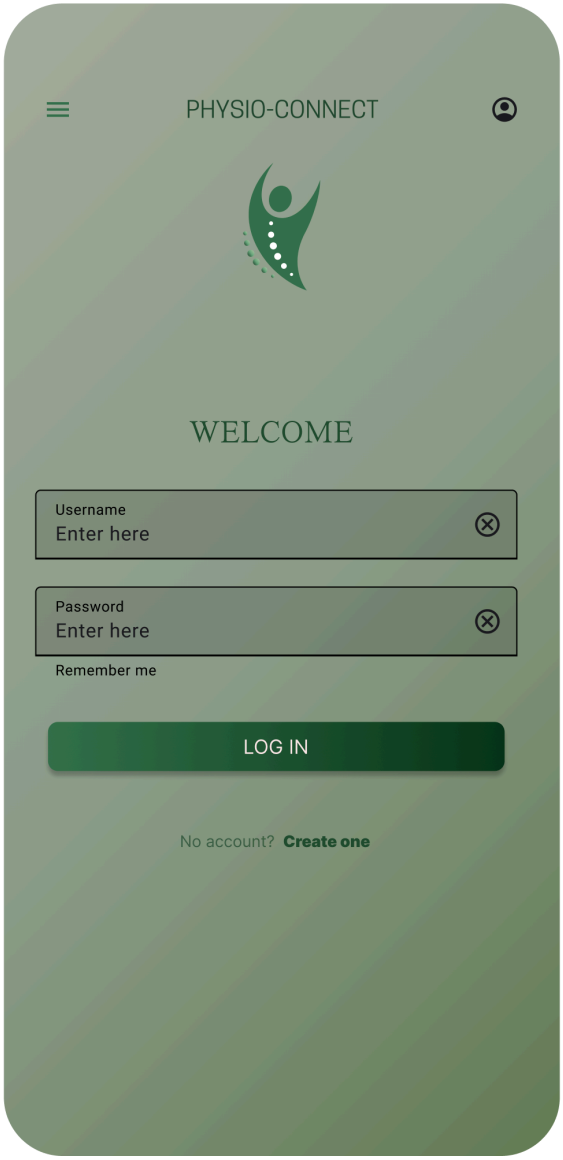
Figure 24: Virtual Consultation (Self Composed)

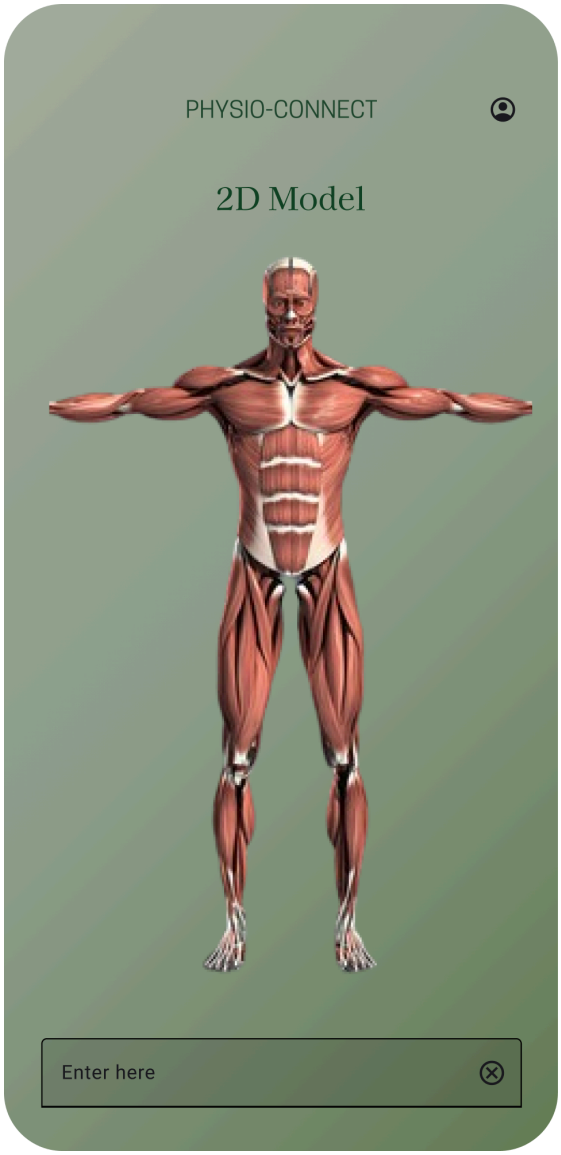
6.3.3.UI Design and mockups

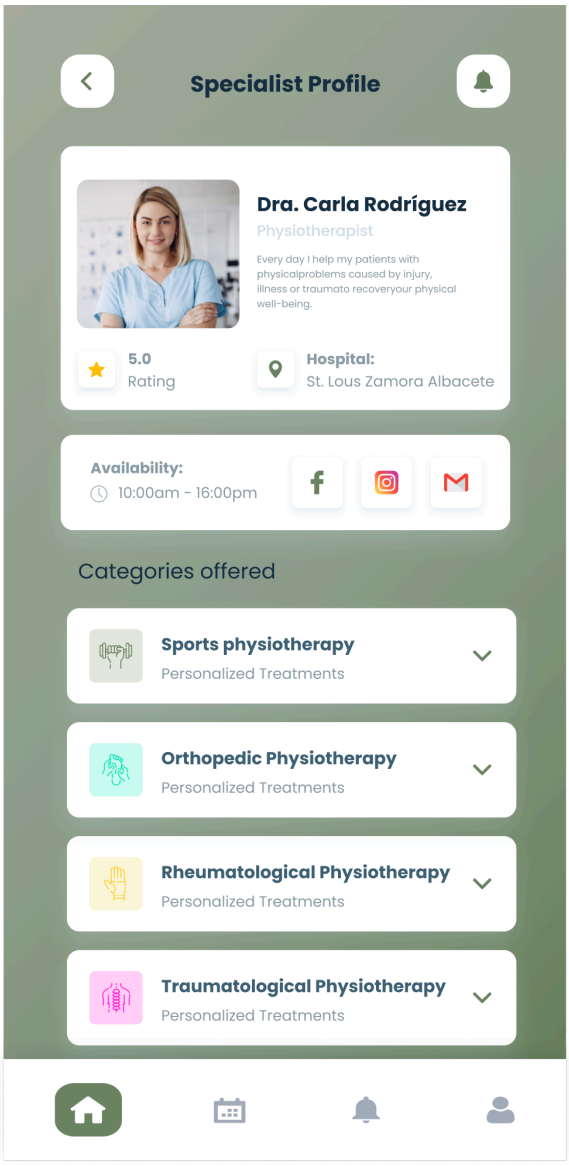
This section features the design prototypes and user interface mockups, providing a visual representation of the app's layout and functionality.

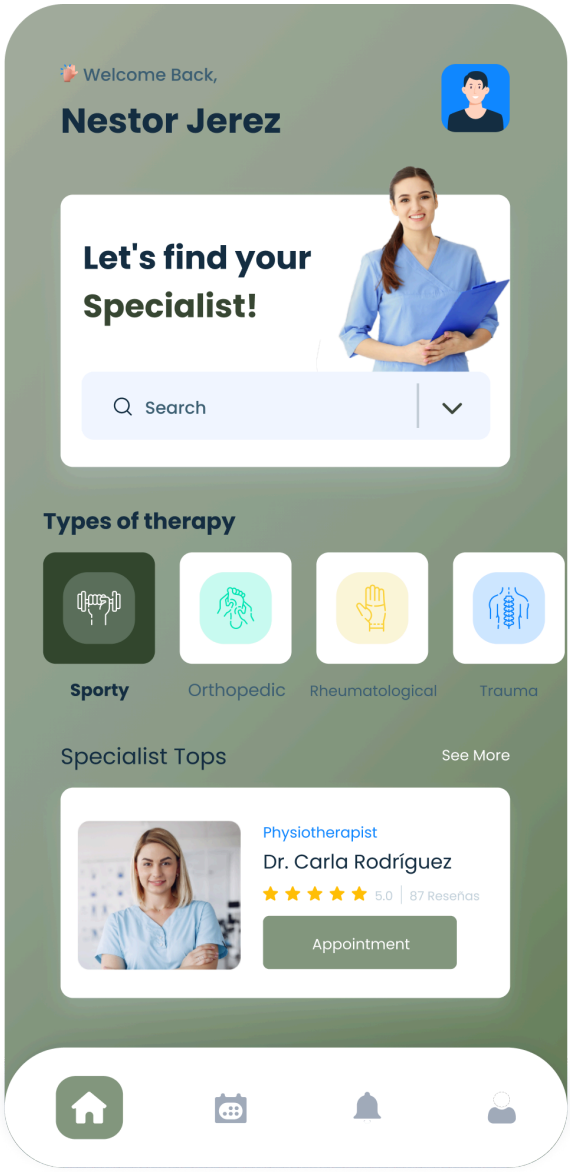




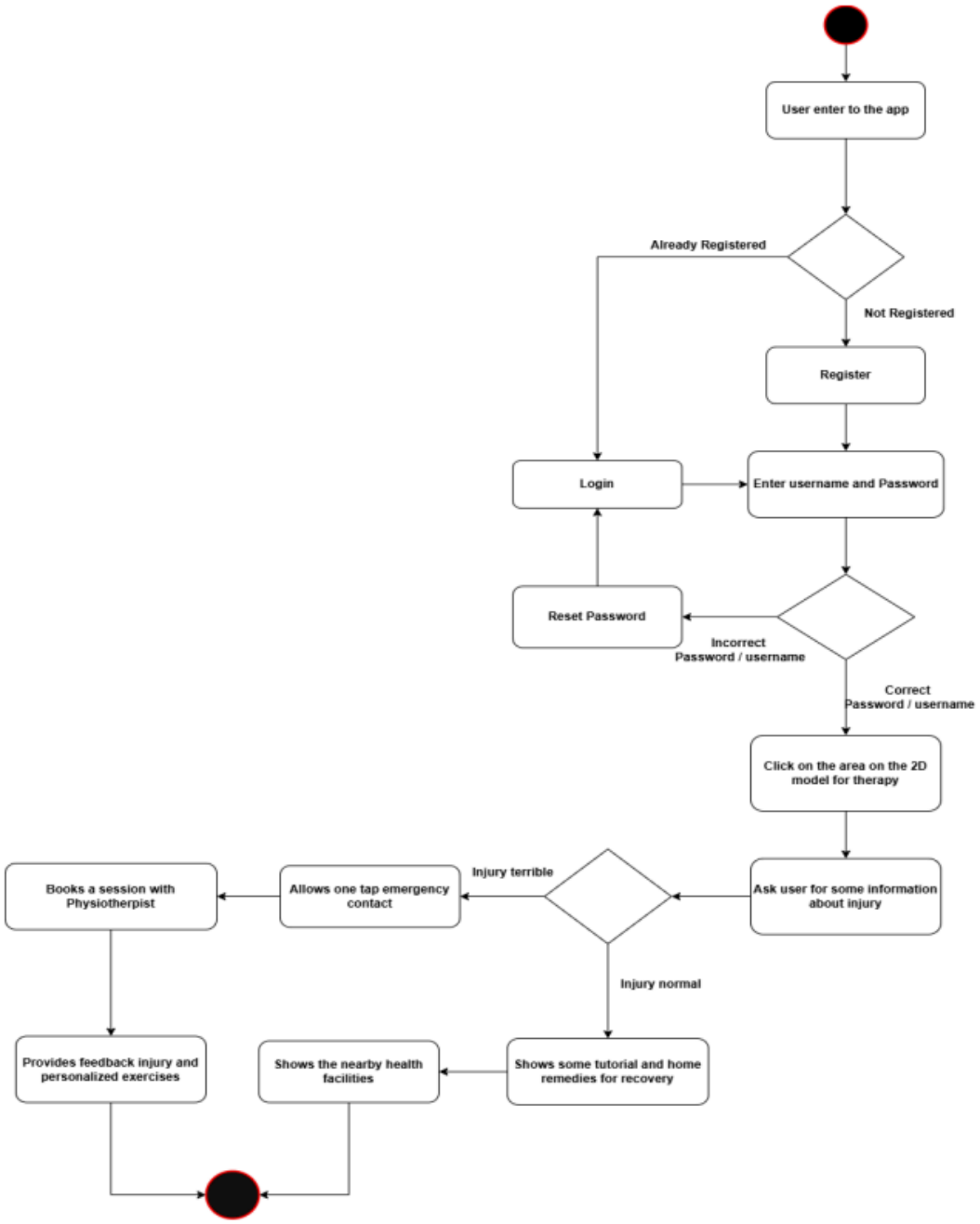

PHYSIO-CONNECT
WELCOME
Username
Enter here
Password
Enter here
Remember me
LOG IN
No account? Create one


PHYSIO-CONNECT
2D Model
Enter here


Specialist Profile
Dra. Carla Rodríguez
Physiotherapist
Every day I help my patients with physicalproblems caused by injury, illness or traumato recoveryour physical well-being.
5.0
Rating
Hospital:
St. Lous Zamora Albacete
Availability:
10:00am - 16:00pm
Categories offered
Sports physiotherapy
Personalized Treatments
Orthopedic Physiotherapy
Personalized Treatments
Rheumatological Physiotherapy
Personalized Treatments
Traumatological Physiotherapy
Personalized Treatments


Welcome Back,
Nestor Jerez
Let's find your Specialist!
Search
Types of therapy
Sporty
Orthopedic
Rheumatological
Trauma
Specialist Tops
See More
Physiotherapist
Dr. Carla Rodríguez
5.0
87 Reseñas
Appointment

6.3.4.Activity Diagram



6.4. Chapter Summary

This chapter delves into the design and architecture of software systems, providing a comprehensive exploration of the foundational elements required for effective system development. It begins with an overview of the system architecture, highlighting the organization and interaction of various components.

Key modeling techniques are then examined, including class diagrams to define system structure, data flow diagrams (DFD) to map the movement of data, and sequence diagrams to illustrate dynamic interactions and communication flows between system entities.

The chapter also focuses on UI design and mockups, showcasing user interface layouts to enhance usability, and activity diagrams, which capture workflows and decision-making processes. Together, these tools form a robust framework for designing, visualizing, and documenting the software system, ensuring clarity and efficiency throughout the development process.

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Appendix

What is your gender? *

- ◯ Male
- ◯ Female

What is your age group? *

- ◯ Under 25
- ◯ 25-34
- ◯ 35-44
- ◯ 45-54
- ◯ 55 and above

In which profession are your currently employed? *

Your answer

Does your profession involve physical activities or extended hours of sitting or standing? *

- [ ] Physical activities
- [ ] Extended hours of sitting
- [ ] Extended hours of standing
- [ ] All
- [ ] None of the above

How many hours per day do you typically sit or stand while doing work?

- ◯ less than 4 hours
- ◯ 4-6 hours
- ◯ 6-8 hours
- ◯ More than 8 hours

Have you ever faced any pains or injuries during your work? *

- ◯ Yes
- ◯ No

How familiar are you with K taping and similar techniques for injury prevention and recovery? *

1	2	3	4	5	6	7	8	9	10
◯	◯	◯	◯	◯	◯	◯	◯	◯	◯

Will you be interested in learning basics physiotherapy techniques to help relieve pain at home with the help of an app? *

- ◯ Yes
- ◯ No
- ◯ Maybe

What features would you find most useful in such app?

Your answer

Would you be interested in setting personal recovery or fitness goals within the app?

- ○ Yes
- ○ No

Submit Clear form